

Machine Learning

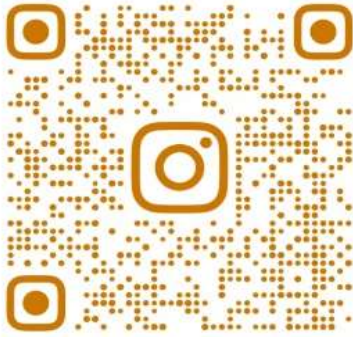
(BE Computer 2019 PAT)

A.Y. 2022-23

Prepared by

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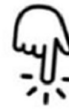


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Unit-2 Feature Engineering

- Concept of Feature, Preprocessing of data: Normalization and Scaling, Standardization, Managing 1Hr
- missing values, Introduction to Dimensionality Reduction, Principal Component Analysis (PCA), 1Hr
- Feature Extraction: Kernel PCA, Local Binary Pattern. 1Hr
- Introduction to various Feature Selection Techniques, Sequential 1Hr
- Forward Selection, Sequential Backward Selection. 1Hr
- Statistical feature engineering: count-based, Length, Mean, Median, Mode etc. based feature vector creation. 1Hr
- Multidimensional Scaling, Matrix Factorization Techniques. 1Hr

Concept of Feature Preprocessing of data:

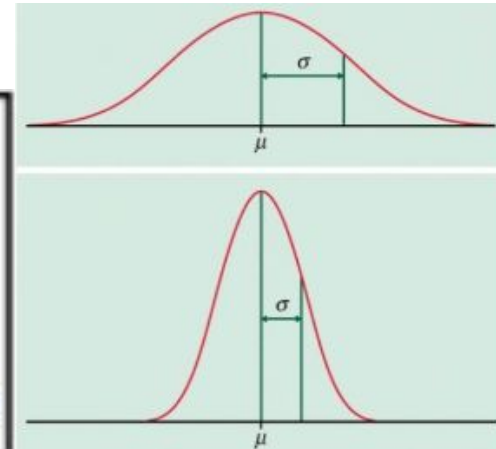
- **Feature Engineering is the process of creating predictive features that can potentially help Machine Learning models achieve a desired performance.** In most of the cases, features will be measurements of different unit and range of values.
- Feature scaling helps decrease the time to find support vectors for SVMs

Diagram illustrating the formula for Feature Scaling (Z-score normalization):

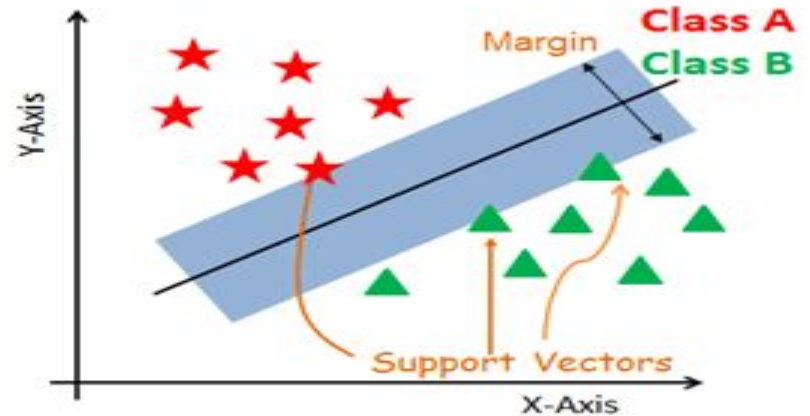
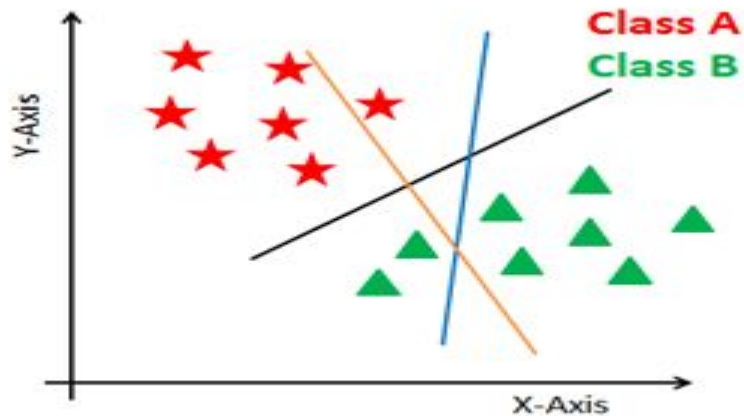
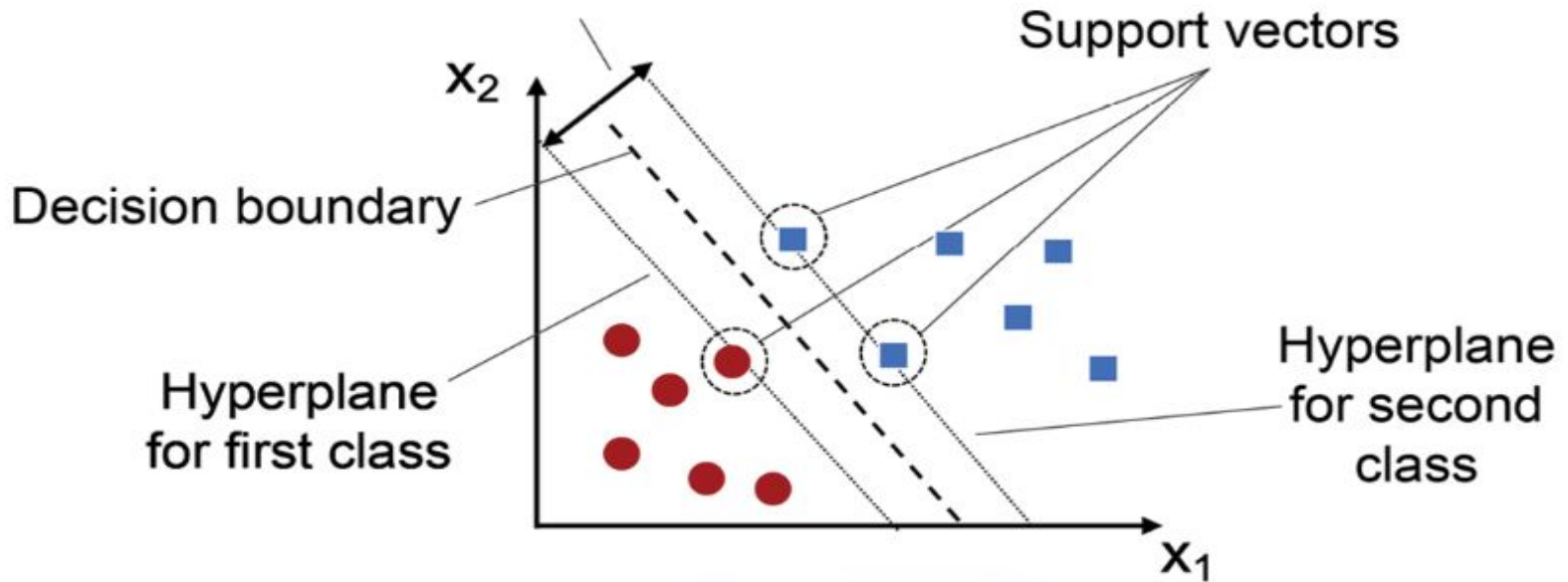
$$X' = \frac{x - \text{mean}(x)}{a}$$

Labels and arrows in the diagram:

- new value** points to X' .
- original value** points to x .
- mean** points to $\text{mean}(x)$.
- Standard deviation** points to a .

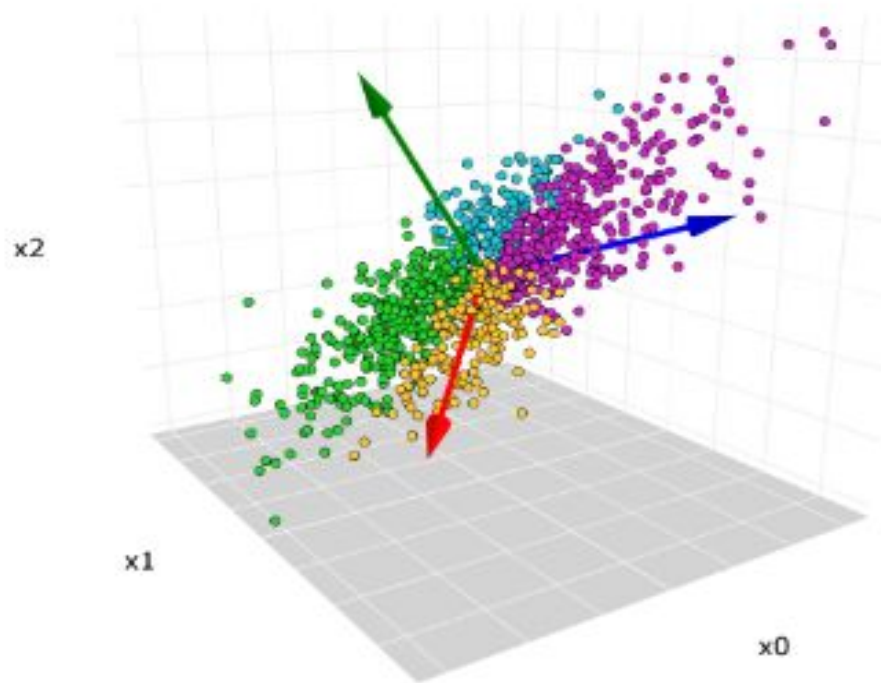


Margin (gap between decision boundary and hyperplanes)

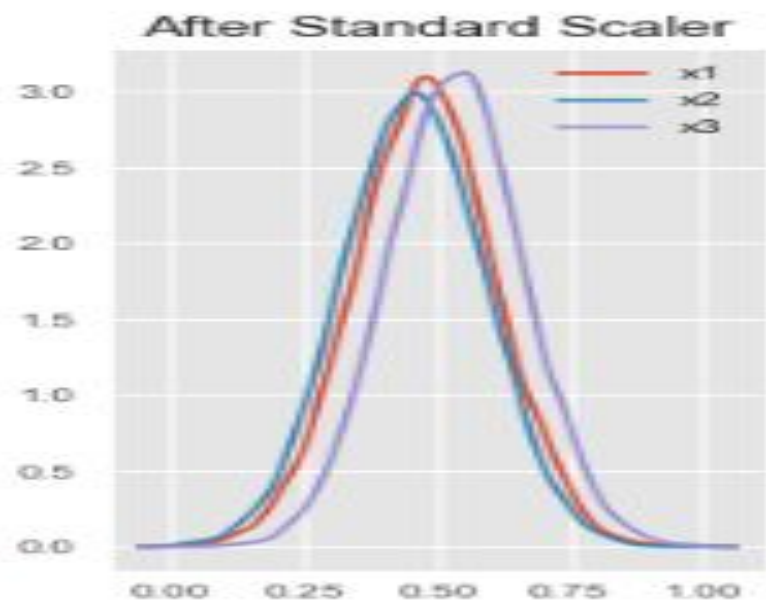
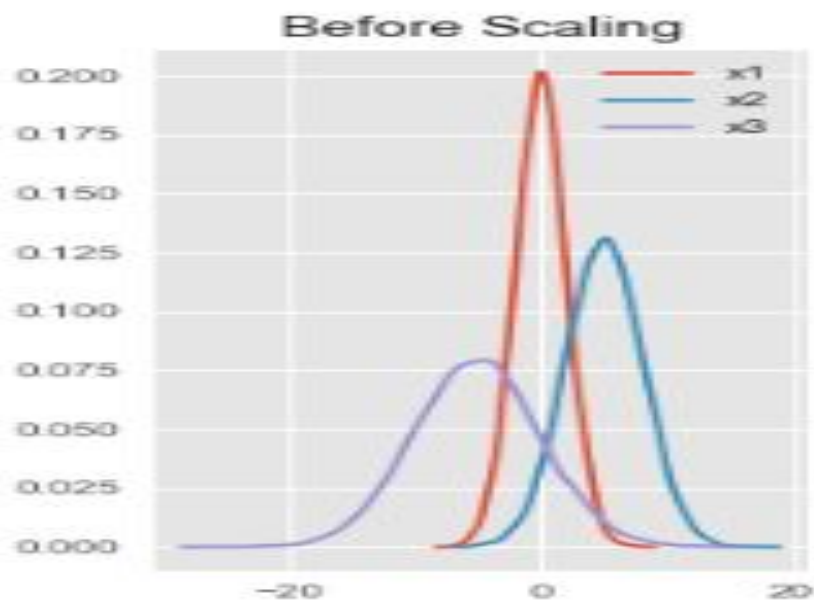
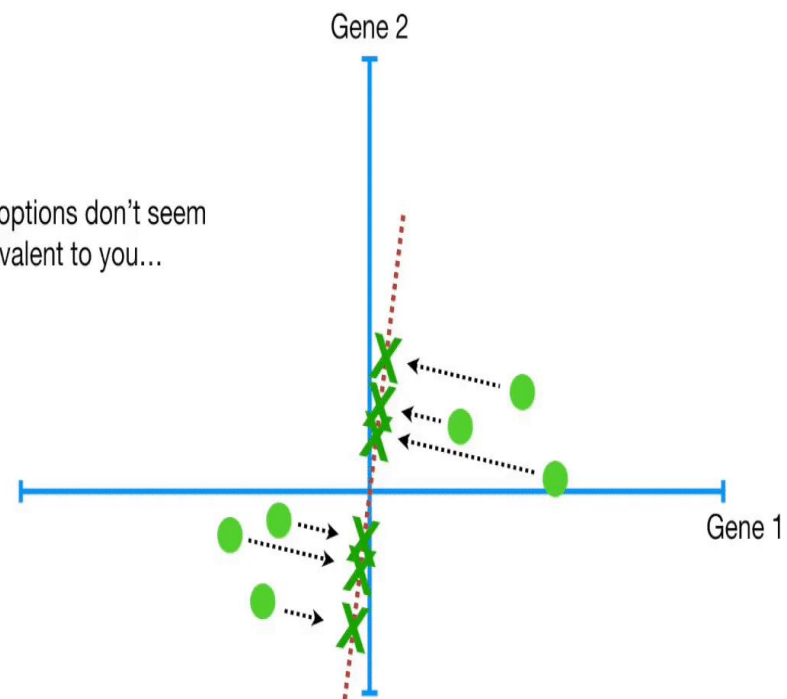


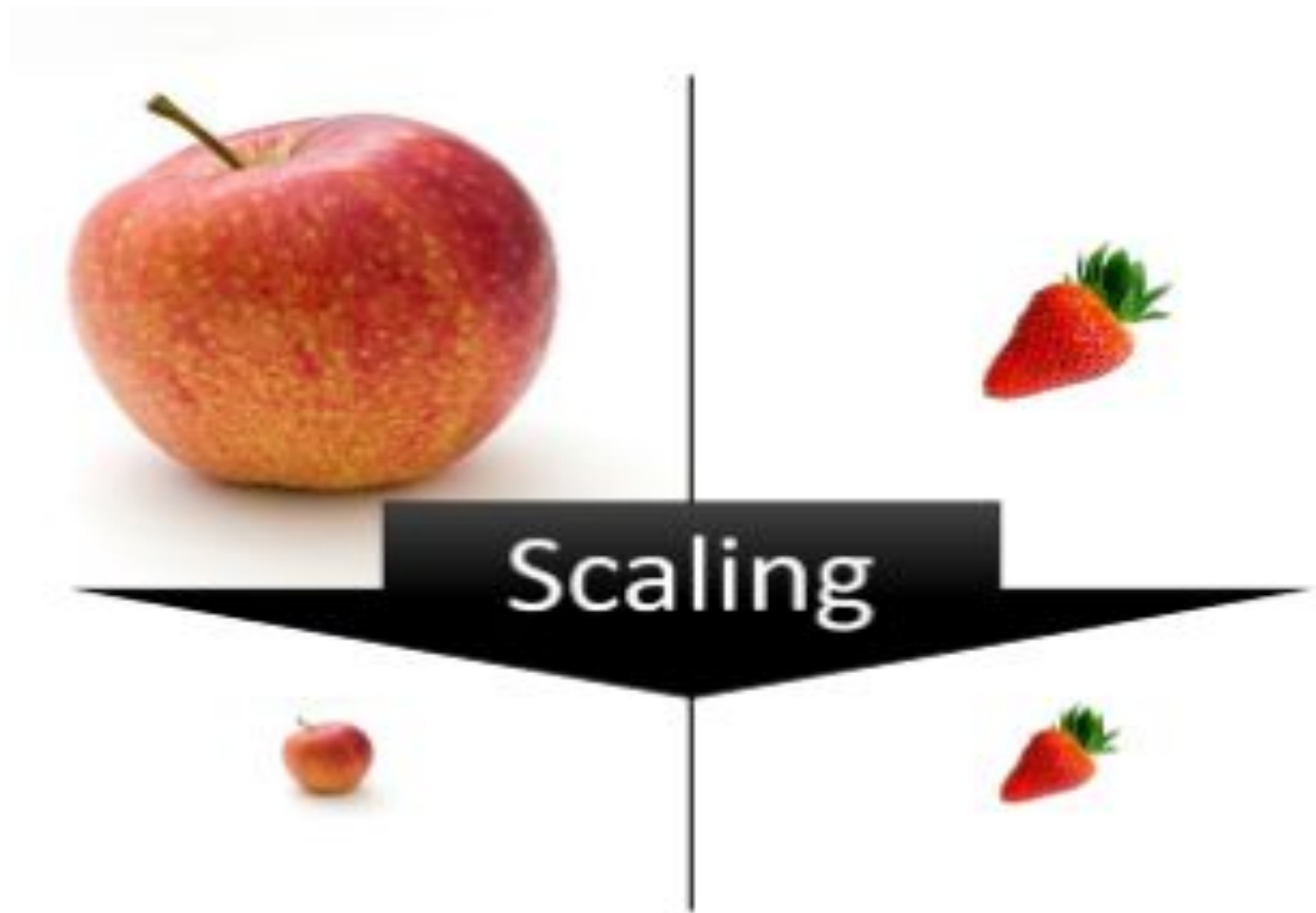
Concept of Feature Preprocessing of data:

- **Having features varying in scale and range could be an issue** when the model we are trying to build uses **distance measures such as Euclidean Distance**. Such models could be **K-Nearest Neighbours, K-Means Clustering**. (Euclidean distances are sensitive to feature magnitude.)
- **Principal Component Analysis (PCA)** is also a good example of when feature scaling is important. PCA require the features to be centered at 0.



If those options don't seem equivalent to you...





Apple & Strawberry

Concept of Feature Preprocessing of data:

- If we want to get the best-mixed juice, we need to mix all fruit not by their size but based on their right proportion.
- We just need to remember apple and strawberry are not the same unless **we make them similar in some context to compare their attribute.**
- Similarly, in many **machine learning algorithms, to bring all features in the same standing, we need to do scaling so that one significant number doesn't impact the model just because of their large magnitude.**

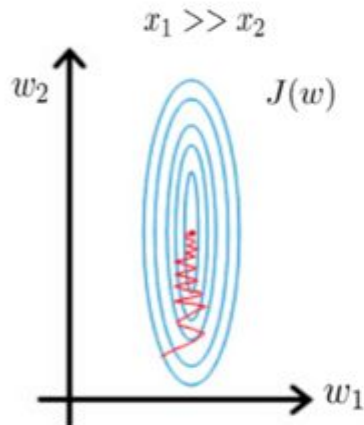
Concept of Feature Preprocessing of data:

- Some examples of algorithms where feature scaling matters are:
- **K-nearest neighbors (KNN)** with a Euclidean distance measure is sensitive to magnitudes and hence should be scaled for all features to weight in equally.
- **K-Means** uses the Euclidean distance measure here feature scaling matters.
- Scaling is critical while performing **Principal Component Analysis(PCA)**. PCA tries to get the features with maximum variance, and the variance is high for high magnitude features and skews the PCA towards high magnitude features.
- Skews- means- change direction position of feature

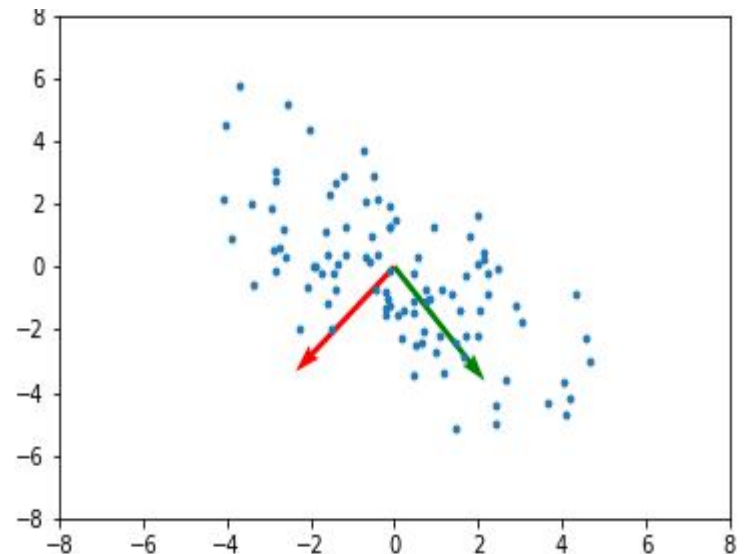
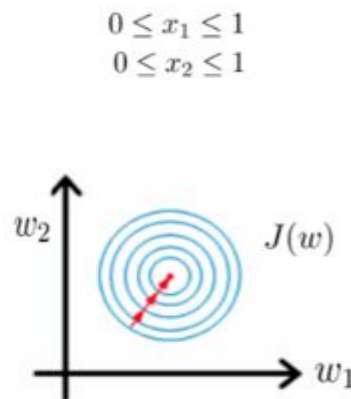
Concept of Feature Preprocessing of data:

- Thus feature scaling is needed to bring every feature in the same footing without any upfront importance.
- if we convert the weight to “Kg,” then “Price” becomes dominant.
- **Eg.** Neural network gradient descent **converge much faster** with feature scaling than without it.

Gradient descent
without scaling



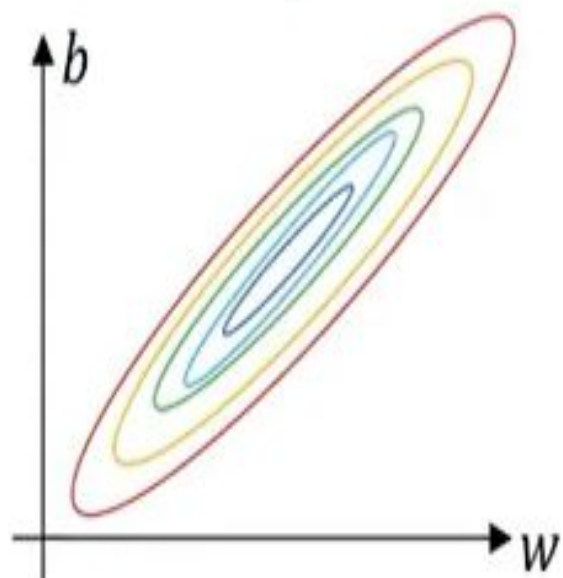
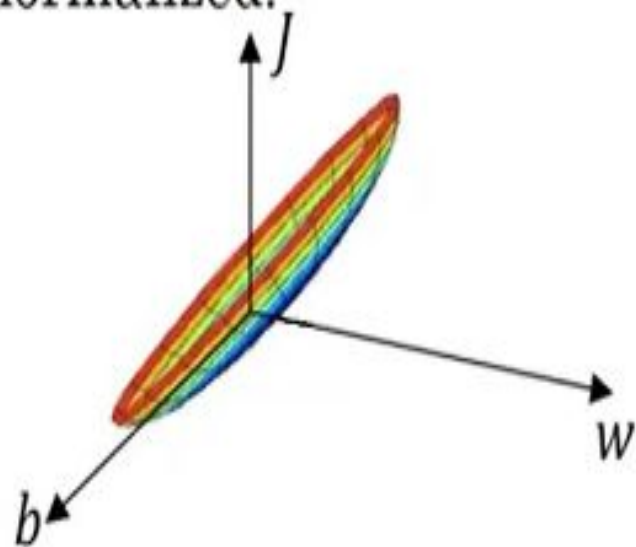
Gradient descent
after scaling variables



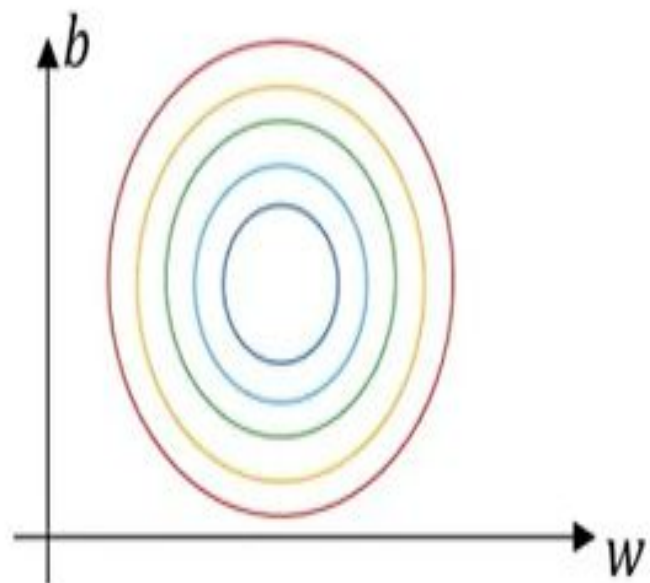
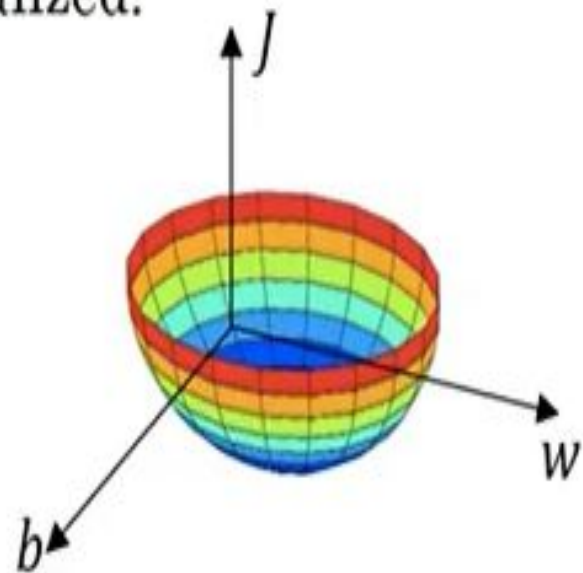
Concept of Feature Preprocessing of data:

- **When we apply Gradient Descent in both situations, it is seen that Gradient Descent converges to the minimum faster if the input is normalized.**
- Whereas, if the input is not normalized, Gradient Descent takes up a lot of steps before it converges to a minimum, which can slow down the learning process of the model.

Unnormalized:



Normalized:



Concept of Feature Preprocessing of data:

- **Feature scaling** in machine learning is one of the most critical steps during the pre-processing of data before creating a machine learning model. **Scaling can make a difference between a weak machine learning model and a better one.**
- The most common techniques of feature scaling are
- **Normalization and Standardization.**

Normalization

- **Normalization** is used when we want to bound our values between two numbers, typically, between $[0,1]$ or $[-1,1]$.
- *which helps us to change the values of numeric columns in the dataset to use a common scale.*
- It is required only when features of machine learning models have different ranges.

Feature scaling



```
graph TD; A[Feature scaling] --> B[Normalization]; A --> C[Standardization]; B --> D["X_new = (X - X_min) / (X_max - X_min)"]; C --> E["X' = (X - Mean) / Standard deviation"];
```

Normalization

$$X_{new} = \frac{X - X_{min}}{X_{max} - X_{min}}$$

Standardization

$$X' = \frac{X - \text{Mean}}{\text{Standard deviation}}$$

Normalization

- Mathematically, we can calculate normalization with the below formula:
- $X_n = (X - X_{\text{minimum}}) / (X_{\text{maximum}} - X_{\text{minimum}})$
- X_n = Value of Normalization
- X_{maximum} = Maximum value of a feature
- X_{minimum} = Minimum value of a feature
- To normalize the machine learning model, **values are shifted and rescaled so their range can vary between 0 and 1. This technique is also known as Min-Max scaling.**
- In this scaling technique, we will change the feature values as follows:

Normalization

Case1- If the value of X is minimum, the value of Numerator will be 0; hence Normalization will also be 0.

$$X_n = (X - X_{\text{minimum}}) / (X_{\text{maximum}} - X_{\text{minimum}})$$

Put $X = X_{\text{minimum}}$ in above formula, we get;

$$X_n = X_{\text{minimum}} - X_{\text{minimum}} / (X_{\text{maximum}} - X_{\text{minimum}})$$

$$X_n = 0$$

Case2- If the value of X is maximum, then the value of the numerator is equal to the denominator; hence Normalization will be 1.

$$X_n = (X - X_{\text{minimum}}) / (X_{\text{maximum}} - X_{\text{minimum}})$$

Put $X = X_{\text{maximum}}$ in above formula, we get;

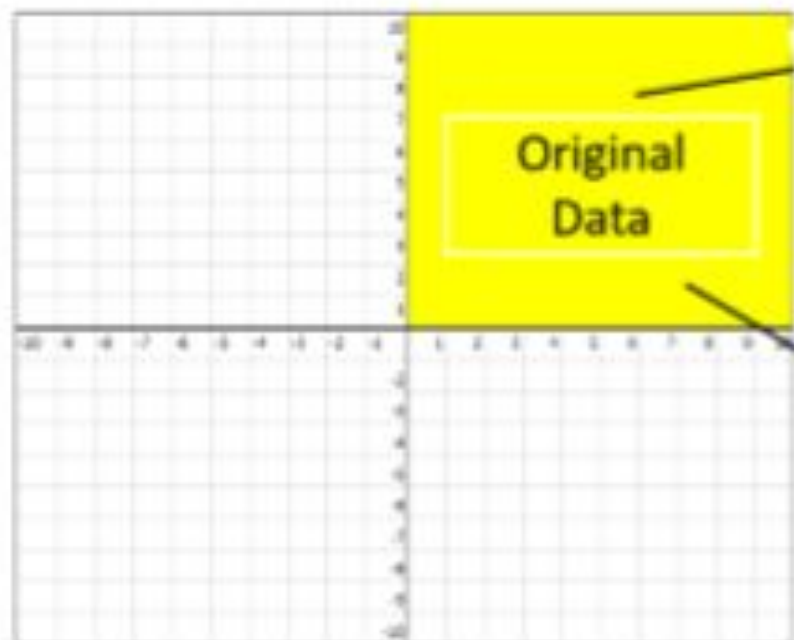
$$X_n = X_{\text{maximum}} - X_{\text{minimum}} / (X_{\text{maximum}} - X_{\text{minimum}})$$

Normalization

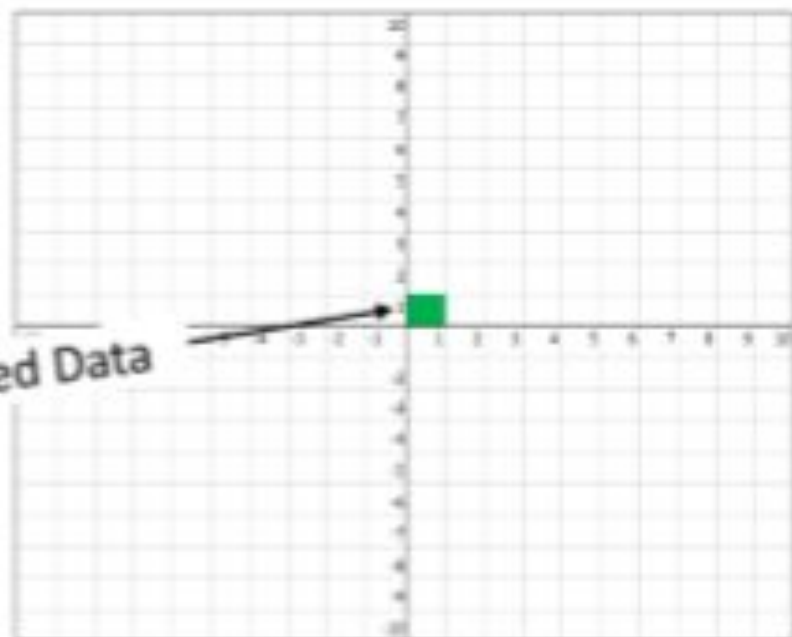
- **Case3-** On the other hand, if the value of X is neither maximum nor minimum, then values of normalization will also be between 0 and 1.
- Hence, Normalization can be defined as a scaling method where values are shifted and rescaled to maintain their ranges between 0 and 1, or in other words; it can be referred to as **Min-Max scaling technique**.

Normalization

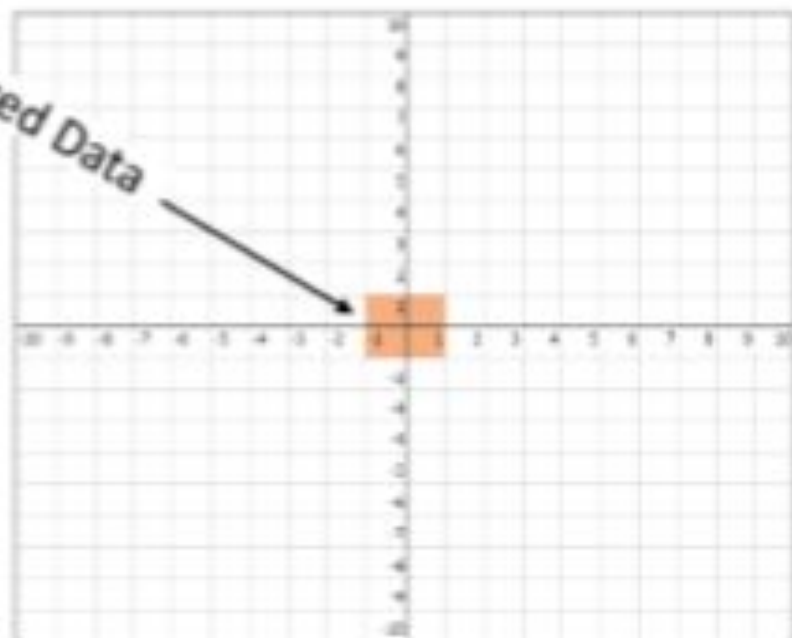
- Normalization techniques in Machine Learning
- **Min-Max Scaling:** This technique is also referred to as scaling. ranging between 0 and 1.
- **Standardization scaling:**
- Standardization scaling is also known as **Z-score** normalization, in which values are centered around the mean with a unit standard deviation, which means the attribute becomes zero and the resultant distribution has a unit standard deviation.



Normalized Data



Standardized Data



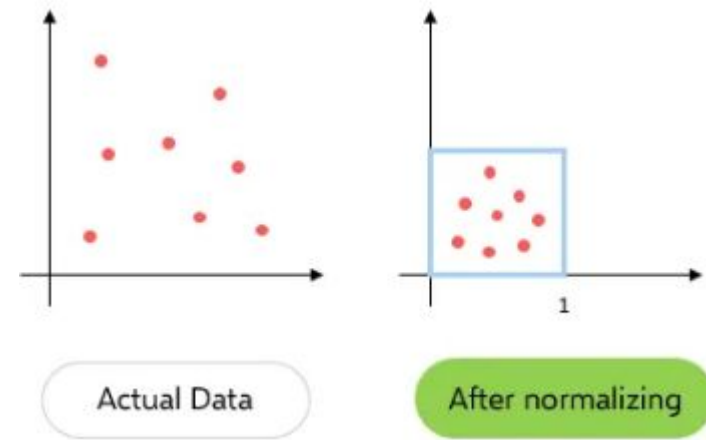


This Z is called standard score and it represents the number of standard deviations above or below the mean that a specific observation falls. i.e. If $Z=2$, it means that the observation lies two standard deviations above the mean. If we plot these standard scores it will be a normal distribution with mean at 0 and the standard deviation equals to 1.

$$Z = \frac{x - \mu}{\sigma}$$

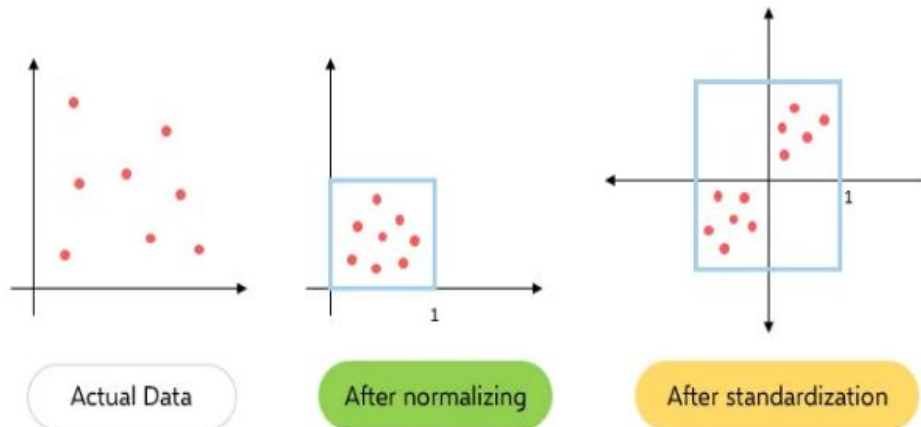
Standardising a variable does not normalize the distribution of the data.

- centers the mean at 0
- scales the variance at 1
- preserves the shape of the original distribution
- the minimum and maximum values of the different variables may vary
- preserves outliers



Scaling to Minimum and Maximum values — MinMaxScaling

- does not center the mean at 0
- variance varies across variables
- may not preserve the shape of the original distribution
- the minimum and maximum values are 0 and 1.
- sensitive outliers



- While **Standardization transforms the data to have zero mean and a variance of 1**, they make our data **unitless**.
- The machine learning algorithm works on **numbers** and does not know what that number represents. A weight of 10 grams and a price of 10 dollars represents completely two different things — which is a no brainer for humans, but **for a model as a feature, it treats both as same**.

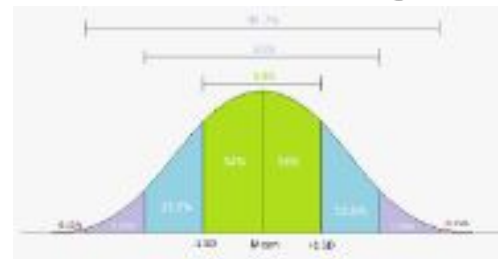
Concept of Feature Preprocessing of data:

- Suppose we have two features of weight and price, as in the below table. The “Weight” cannot have a meaningful comparison with the “Price.” So the assumption algorithm makes that since “Weight” > “Price,” thus “Weight,” is more important than “Price.”

Name	Weight	Price
Orange	15	1
Apple	18	3
Banana	12	2
Grape	10	5

Standardization scaling

- **Standard Deviation Definitions**
- The standard deviation (SD) is a quantification that **measures the distribution (dispersion) of the data set relative to its mean.**
- It is calculated as the square root of the variance.
- It is denoted by the lower Greek letter σ (**sigma**).
- If the deviation is greater, the dispersion will be greater.
- **if the deviation is lesser greater the uniformity.**

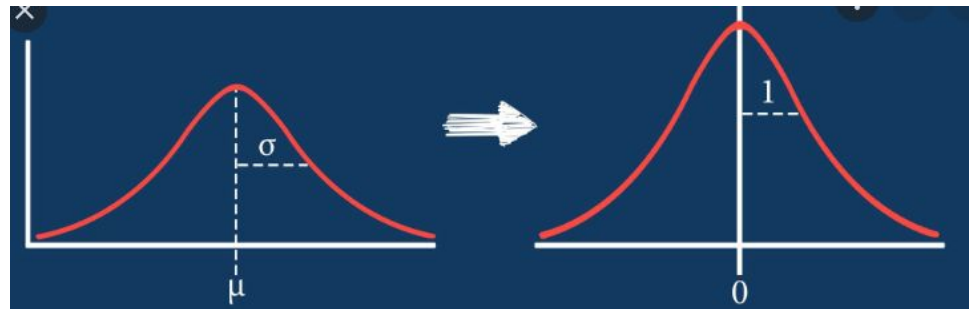


Standardization

- Hence, standardization can be expressed as follows:

$$X' = \frac{X - \mu}{\sigma}$$

- Here, μ represents the mean of feature value, and σ represents the standard deviation of feature values.
- However, unlike Min-Max scaling technique, feature values are not restricted to a specific range in the standardization technique.



Standardization scaling

- The standard deviation is the **measure of the variations** of all values from the mean.
- Standard deviation is the square root of the sum of squared deviation from the mean divided by the number of observations.
- It is the square root of the **variance**.
- **Variance**
- It defines how much a random variable differs from its expected value.
- It is the average of the squares of the differences between expected and individual value.
- It can never have a negative value.
- It is denoted by σ^2 .

Difference between Normalization and Standardization

Normalization	Standardization
This technique uses minimum and max values for scaling of model.	This technique uses mean and standard deviation for scaling of model.
It is helpful when features are of different scales.	It is helpful when the mean of a variable is set to 0 and the standard deviation is set to 1.
Scales values ranges between [0, 1] or [-1, 1].	Scale values are not restricted to a specific range.
It got affected by outliers.	It is comparatively less affected by outliers.
Scikit-Learn provides a transformer called MinMaxScaler for Normalization.	Scikit-Learn provides a transformer called StandardScaler for Normalization.
It is also called Scaling normalization.	It is known as Z-score normalization.
It is useful when feature distribution is unknown.	It is useful when feature distribution is normal.

When to use Normalization or Standardization?

- Feature distribution of data does not follow a **Gaussian (bell curve)** distribution. Normalization must have an abounding range, so **if you have outliers in data, they will be affected by Normalization.**
- it is also useful for data having variable scaling techniques such as **KNN, artificial neural networks.**
- Hence, you can't use assumptions for the distribution of data.

When to use Normalization or Standardization?

- Standardization in the machine learning model is useful when your data follows a Gaussian distribution. However, this does not have to be necessarily true.
- Standardization does not necessarily have a bounding range, so **if you have outliers in your data, they will not be affected by Standardization.**
- Further, it is also useful when data has variable dimensions and techniques such as **linear regression, logistic regression, and linear discriminant analysis.**

Missing Data Values And How To Handle It



Managing Missing Values

Missing Completely At Random (MCAR)-

- Missing values are completely independent of other data. There is no pattern.
- In the case of MCAR, the data could be missing due to human error, some system/equipment failure, loss of sample, or some unsatisfactory technicalities while recording the values.
- **there is no relationship between the missing data and any other values observed or unobserved**
- For Example, suppose in a library there are some overdue books. Some values of overdue books in the computer system are missing. The reason might be a human error like the librarian forgot to type in the values. So, the missing values of overdue books are not related to any other variable/data in the system.

Managing Missing Values

Missing At Random (MAR)-

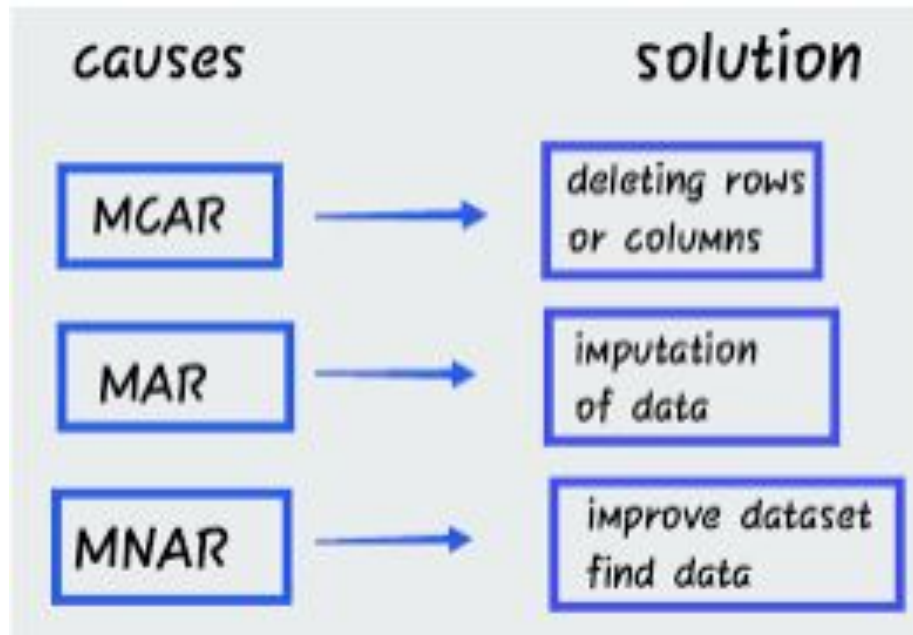
- Missing at random (MAR) means **there is some relationship between the missing data and other values/data.**
- **Eg.** Suppose a poll is taken for overdue books of a library. Gender and the number of overdue books are asked in the poll. Assume that most of the females answer the poll and men are less likely to answer. So why the data is missing can be explained by another factor that is gender.

Managing Missing Values

- **Missing Not At Random (MNAR)**
- Missing values depend on the unobserved data.
- If there is some structure/pattern in missing data and other observed data **can not explain** it, then it is Missing Not At Random (MNAR).
- For example, suppose the name and the number of overdue books are asked in the poll for a library. So most of the people having no overdue books are likely to answer the poll. People having more overdue books are less likely to answer the poll.

How To Handle The Missing Data

- Deleting the Missing values- Row/Column
- Imputing the Missing Values- **Replacing With Arbitrary Value, Mean Value, Mode Value, Median Value**
- Impute the Most Frequent Value.



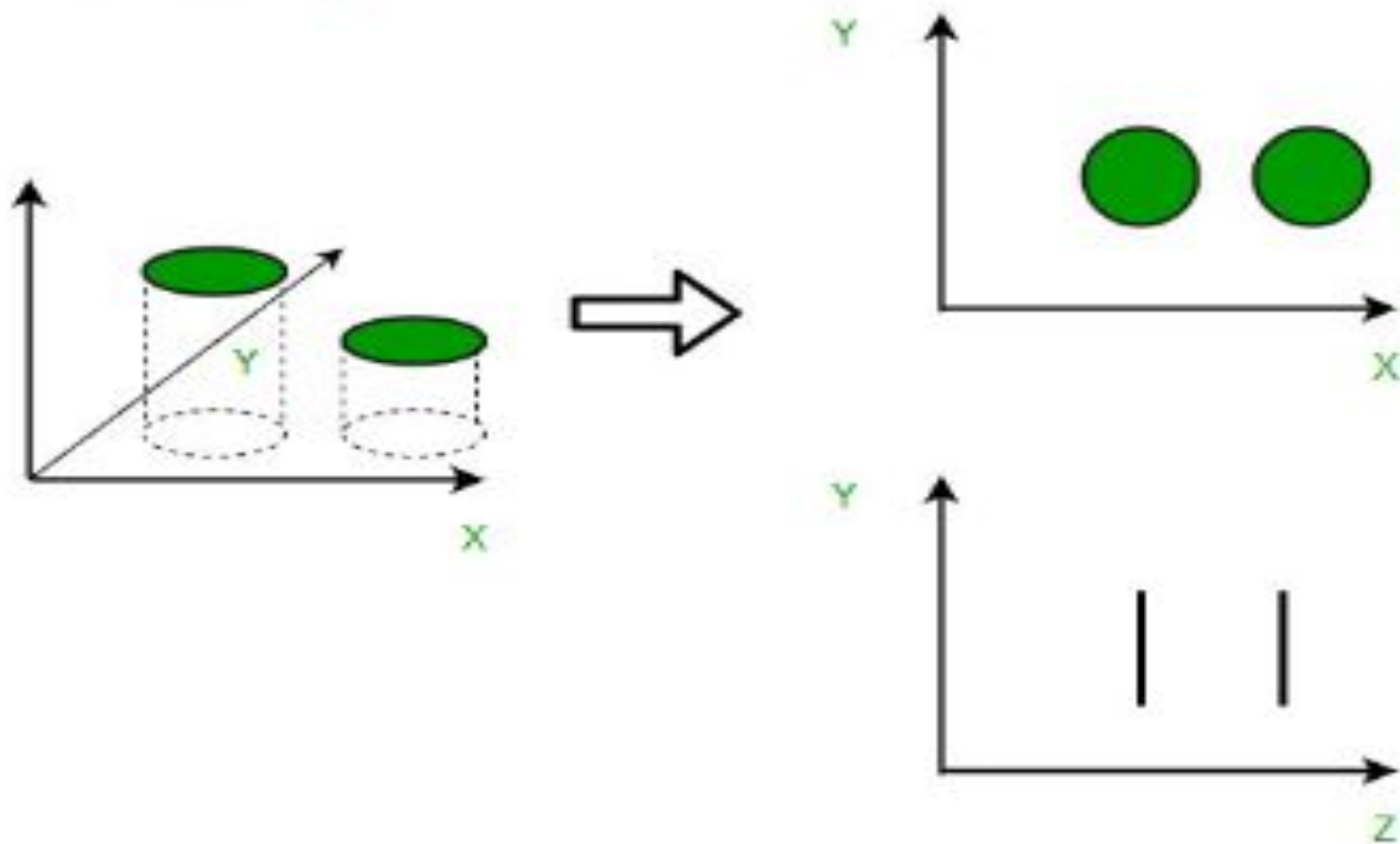
Introduction to Dimensionality Reduction

- What is Dimensionality Reduction?
- The number of input features, variables, or columns present in a given dataset is known as **Dimensionality**, and the process to reduce these features is called dimensionality reduction.
- *"It is a way of converting the higher dimensions dataset into lesser dimensions dataset ensuring that it provides similar information."*

Introduction to Dimensionality Reduction

- It is used to solving the classification and regression problems.
- It is commonly used in the fields that deal with high-dimensional data, such as **speech recognition, signal processing, bioinformatics, etc.** It can also be used for **data visualization, noise reduction, cluster analysis, etc.**
- A 3-D classification problem can be hard to visualize, whereas a 2-D one can be mapped to a simple 2 dimensional space, and a 1-D problem to a simple line. The below figure illustrates this concept, where a 3-D feature space is split into two 1-D feature spaces, and later, if found to be correlated, the number of features can be reduced even further.

Dimensionality Reduction



Components of Dimensionality Reduction

Dimensionality reduction Techniques

```
graph TD; A[Dimensionality reduction Techniques] --> B[Feature Selection]; A --> C[Dimensionality Reduction]; B --> D["• Missing Value Ratio<br>• Low Variance Filter<br>• High Correlation Filter<br>• Random Forest<br>• Backward Feature Extraction<br>• Forward Feature Selection"]; C --> E[Components/Factors based]; C --> F[Projection Based]; E --> G["• Factor Analysis<br>• Principal Component Analysis<br>• Independent Component Analysis"]; F --> H["• ISOMAP<br>• t-SNE<br>• UMAP"];
```

Feature Selection

- Missing Value Ratio
- Low Variance Filter
- High Correlation Filter
- Random Forest
- Backward Feature Extraction
- Forward Feature Selection

Dimensionality Reduction

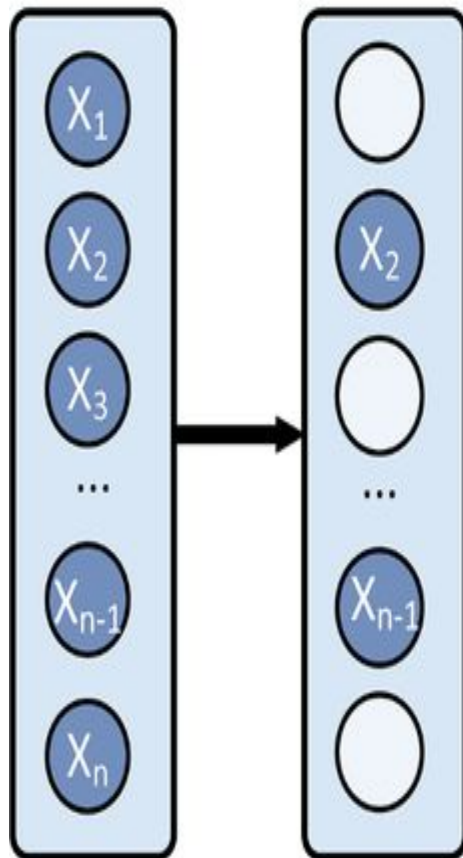
Components/Factors based

- Factor Analysis
- Principal Component Analysis
- Independent Component Analysis

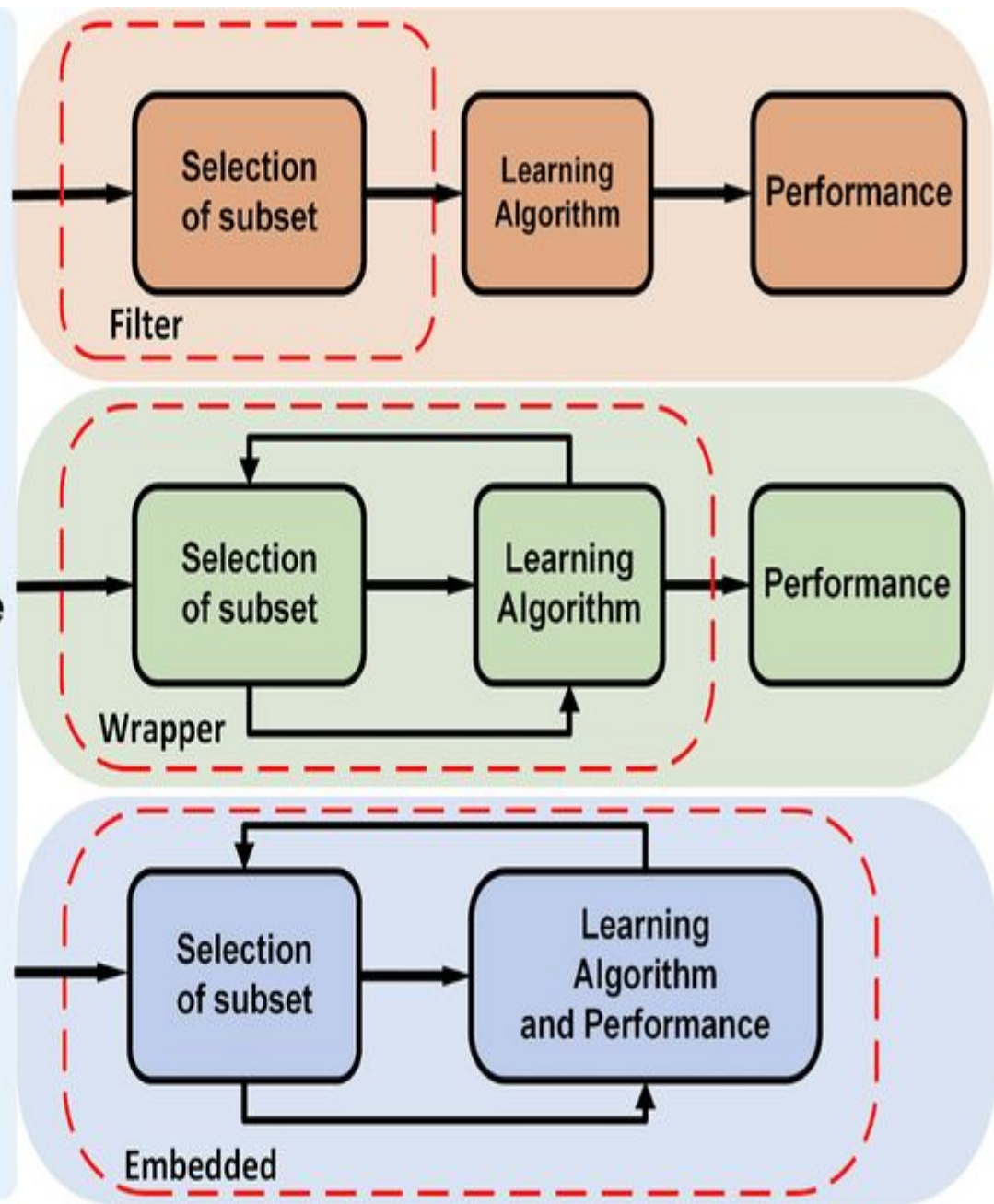
Projection Based

- ISOMAP
- t-SNE
- UMAP

Feature Selection



Feature
Sets



Filter methods	Wrapper methods	Embedded methods
Generic set of methods which do not incorporate a specific machine learning algorithm .	Evaluates on a specific machine learning algorithm to find optimal features.	Embeds (fix) features during model building process . Feature selection is done by observing each iteration of model training phase.
Much faster compared to Wrapper methods in terms of time complexity	High computation time for a dataset with many features	Sits between Filter methods and Wrapper methods in terms of time complexity
Less prone to over-fitting	High chances of over-fitting because it involves training of machine learning models with different combination of features	Generally used to reduce over-fitting by penalizing the coefficients of a model being too large.
Examples – Correlation, Chi-Square test, ANOVA, Information gain etc.	Examples - Forward Selection, Backward elimination, Stepwise selection etc.	Examples - LASSO, Elastic Net, Ridge Regression etc.

- There are two components of dimensionality reduction:
- **Feature selection:** In this, we **try to find a subset of the original** set of variables, or features, to get a smaller subset which can be used to model the problem. It usually involves three ways:
 - Filter
 - Wrapper
 - Embedded
- **Feature extraction:** This reduces the data in a high dimensional space to a lower dimension space, i.e. a space with lesser no. of dimensions.

Methods of Dimensionality Reduction

- The various methods used for dimensionality reduction include:
- **Principal Component Analysis (PCA)**
- **Linear Discriminant Analysis (LDA)**
- **Generalized Discriminant Analysis (GDA)**
- Dimensionality reduction may be both linear or non-linear, depending upon the method used.
- The prime linear method, called Principal Component Analysis, or PCA, is discussed below.

Benefits of applying Dimensionality Reduction

- By reducing the dimensions of the features, the space required to **store the dataset also gets reduced.**
- **Less Computation training time** is required for reduced dimensions of features.
- **Reduced dimensions of features** of the dataset help in visualizing the data quickly.
- **It removes the redundant features** (if present) by taking care of multicollinearity.

Disadvantages of dimensionality Reduction

- Some data may be lost due to dimensionality reduction.
- In the PCA dimensionality reduction technique, sometimes the principal components required to consider are unknown.

Feature Selection

- Feature selection is the process of selecting the
- subset of the relevant features and leaving out the irrelevant features present in a dataset to build a model of high accuracy.
- Three methods are used for the feature selection:

1. Filters Methods

- In this method, the dataset is filtered, and a subset that contains only the relevant features is taken.
- *Some common techniques of filters method are:*
- Correlation
- Chi-Square Test
- ANOVA
- Information Gain, etc.

Feature Selection

2. Wrappers Methods

- In this method, **some features are fed to the ML model, and evaluate the performance.**
- The performance decides whether **to add those features or remove to increase the accuracy of the model.** This method is more accurate than the filtering method but complex to work.
- Some common techniques of wrapper methods are:
 - **Forward Selection**
 - **Backward Selection**
 - **Bi-directional Elimination**

Feature Selection

- **3.Embedded Methods:** Embedded methods check the different training iterations of the machine learning model and evaluate the importance of each feature.
- Some common techniques of Embedded methods are:
- **LASSO**
- **Elastic Net**
- **Ridge Regression, etc.**

Feature Extraction:

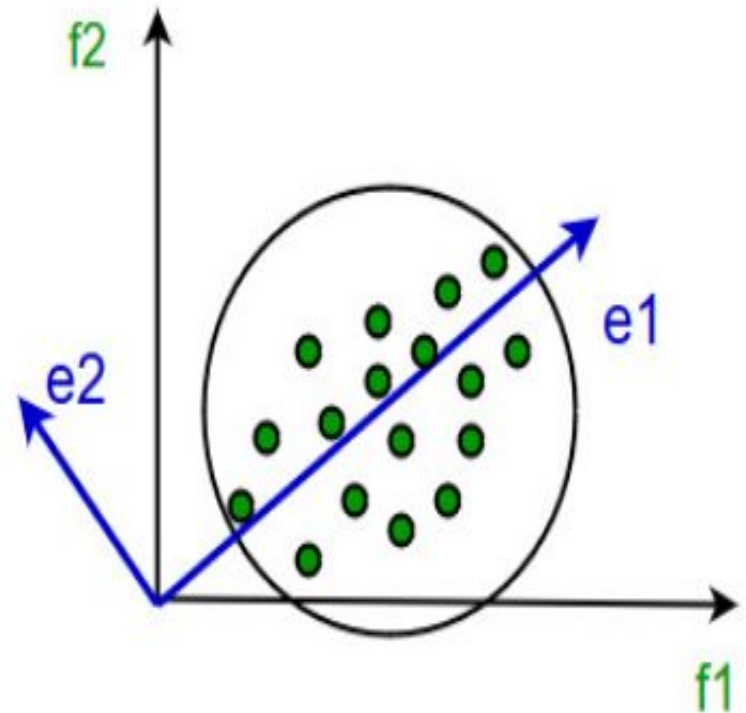
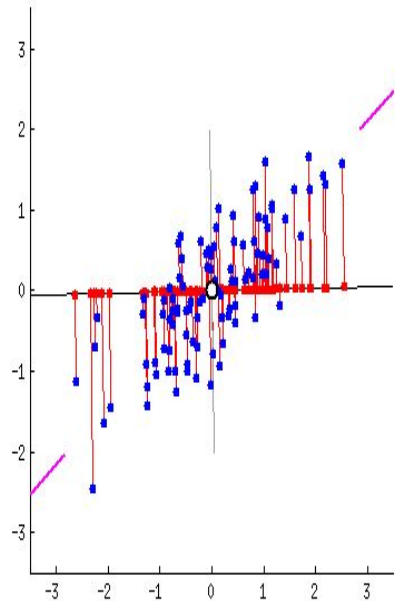
- Feature extraction is the process of transforming the space containing many dimensions into space with fewer dimensions.
- This approach is useful when we want to keep the whole information but use fewer resources while processing the information.
- Some common feature extraction techniques are:
 - **Principal Component Analysis**
 - **Linear Discriminant Analysis**
 - **Kernel PCA**
 - **Quadratic Discriminant Analysis**

Common techniques of Dimensionality Reduction

- **Principal Component Analysis**
- **Backward Elimination**
- **Forward Selection**
- **Score comparison**
- **Missing Value Ratio**
- **Low Variance Filter**
- **High Correlation Filter**
- **Random Forest**
- **Factor Analysis**
- **Auto-Encoder**

Principal Component Analysis

- This method was introduced by Karl Pearson. It works on a condition that **while the data in a higher dimensional space is mapped to data in a lower dimension space, the variance of the data in the lower dimensional space should be maximum.**



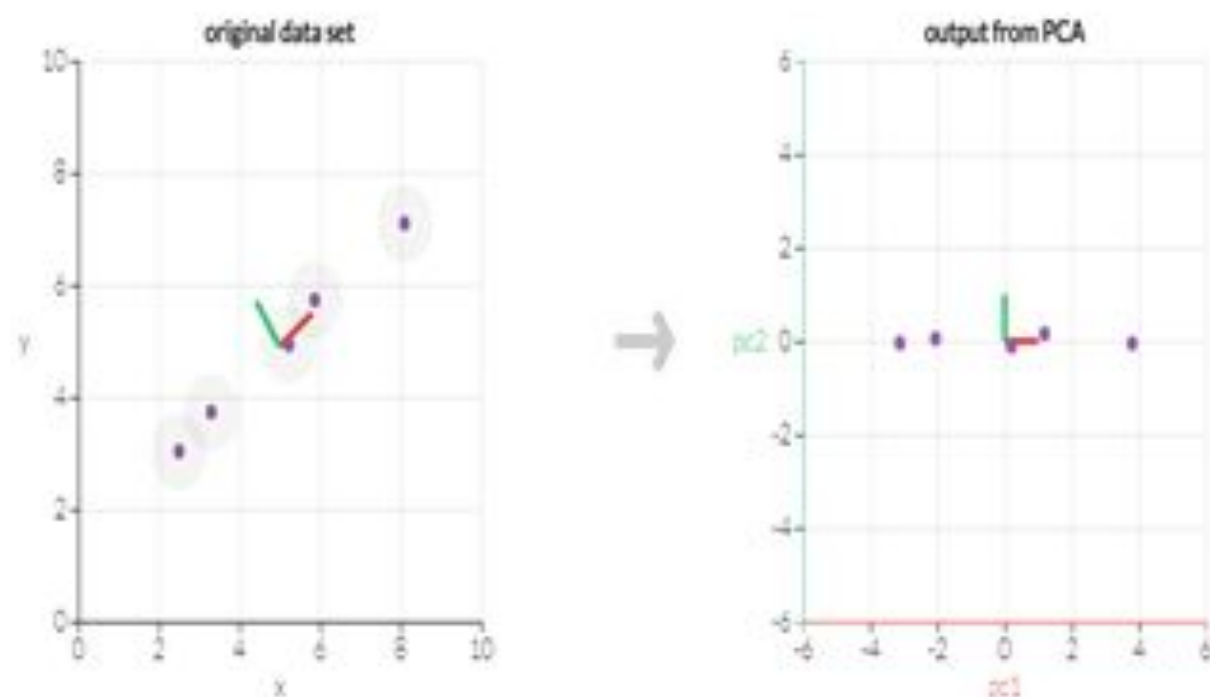
Principal Component Analysis

- Principal Component Analysis is a statistical process that **converts the observations of correlated features into a set of linearly uncorrelated features** with the help of orthogonal transformation. **These new transformed features are called the Principal Components.**
- It is one of the popular tools that is used for exploratory data analysis and predictive modeling.

Principal Component Analysis

- PCA works by considering the variance of each attribute because the high attribute shows the good split between the classes, and hence it reduces the dimensionality.
- Some real-world applications of PCA are *image processing, movie recommendation system, optimizing the power allocation in various communication channels.*

Drag the points around in the following visualization to see PC coordinate system adjusts.



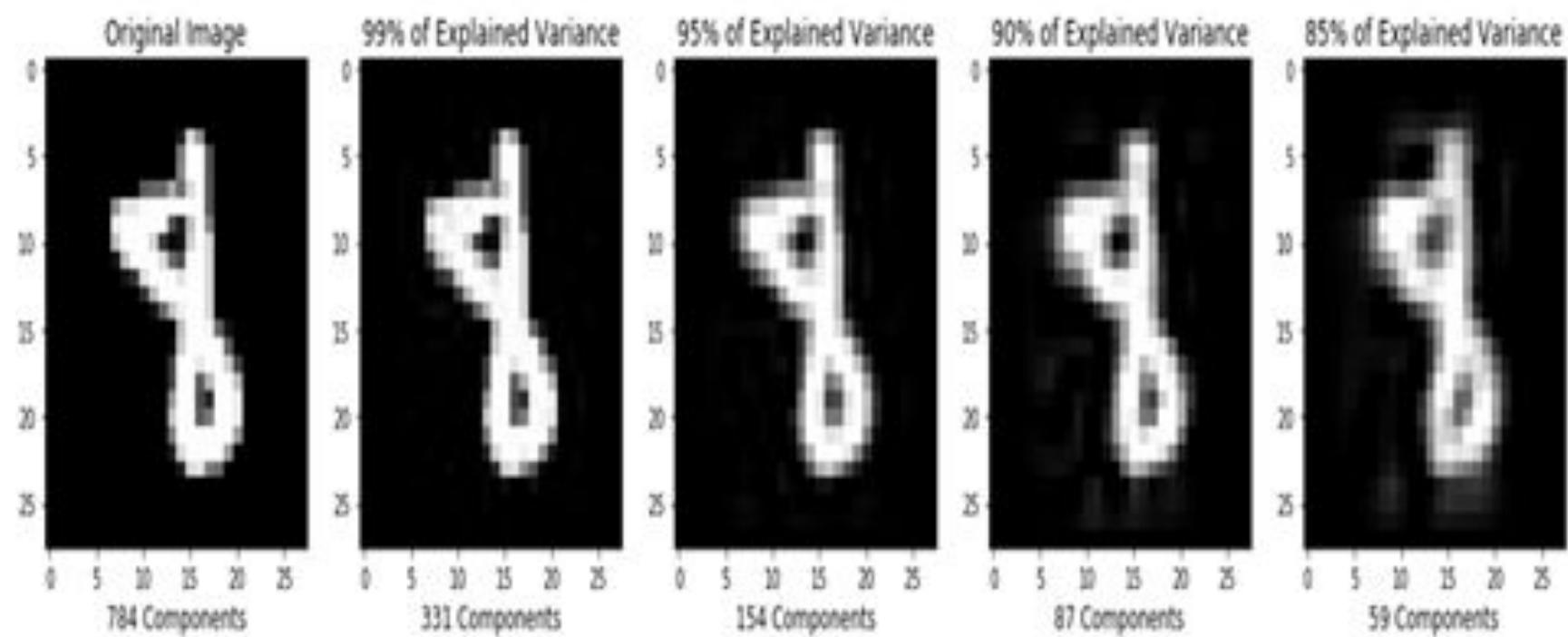
PCA is useful for eliminating dimensions. Below, we've plotted the data along a pair of lines: one composed of the x -values and another of the y -values.

If we're going to only see the data along one dimension, though, it might be better to make that dimension the principal component with most variation. We don't lose much by dropping **PC2** since it contributes the least to the variation in the data set.

In three dimensions, PCA is more useful, because it's hard to see through a cloud of data. In the example below, the original data is plotted in 3D, but you can project the data into 2D through a transformation no different than finding a camera angle: rotate the axes to find the best angle. To see the "official" PCA transformation, click the "Show PCA" button. The PCA transformation ensures that the horizontal axis PC1 has the most variation, the vertical axis PC2 the second-most, and a third axis PC3 the least. Obviously PC3 is the one we drop.



ting in the UK (a 17D example)



Principal Component Analysis

- The PCA algorithm is based on some mathematical concepts such as:
- **Variance and Covariance**
- **Eigenvalues and Eigen factors**

Principal Component Analysis

- **Variance** refers to the **spread of a data set around its mean value.**
- It is calculated by finding the probability-weighted average of squared deviations from the expected value.
- While performing market research, variance is particularly useful when calculating probabilities of future events.
- Variance is a great way to find all of the possible values and likelihoods that a random variable can take within a given range.

Principal Component Analysis

- A variance value of zero represents that all of the values within a data set are identical, while all variances that are not equal to zero will come in the form of positive numbers.
- The larger the variance, the more spread in the data set.
- A large variance means that the numbers in a set are far from the mean and each other
- A small variance means that the numbers are closer together in value.

Principal Component Analysis

- **X** represents an individual data point,
- **u** represents the mean of the data points,
- **N** represents the total number of data points.
- Note that while calculating a sample variance in order to estimate a population variance, the denominator of the variance equation becomes $N - 1$.
- This removes bias from the estimation, as it prohibits the researcher from underestimating the population variance.

$$\sigma^2 = \frac{\Sigma (x - \mu)^2}{N}$$

Principal Component Analysis

An Advantage of Variance

- One of the primary advantages of variance is that it treats all deviations from the mean of the data set in the same way, regardless of direction.
- **A Disadvantage of Variance**
- it gives added weight to numbers that are far from the mean, or outliers.
- Squaring these numbers can at times result in skewed interpretations of the data set as a whole.

Principal Component Analysis

- **Covariance** refers to the **measure of the directional relationship between two random variable.**
- **A positive covariance** means that the two variables at hand are **positively related**, and they move in the same direction.
- **A negative covariance** means that the variables are **inversely related**, or that they move in opposite directions.

Principal Component Analysis

- **X** represents the independent variable,
- **Y** represents the dependent variable,
- **N** represents the number of data points in the sample,
- **x-bar** represents the mean of the **X**, and
- **y-bar** represents the mean of the dependent variable **Y**.

$$COV(x, y) = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{n - 1}$$

Principal Component Analysis

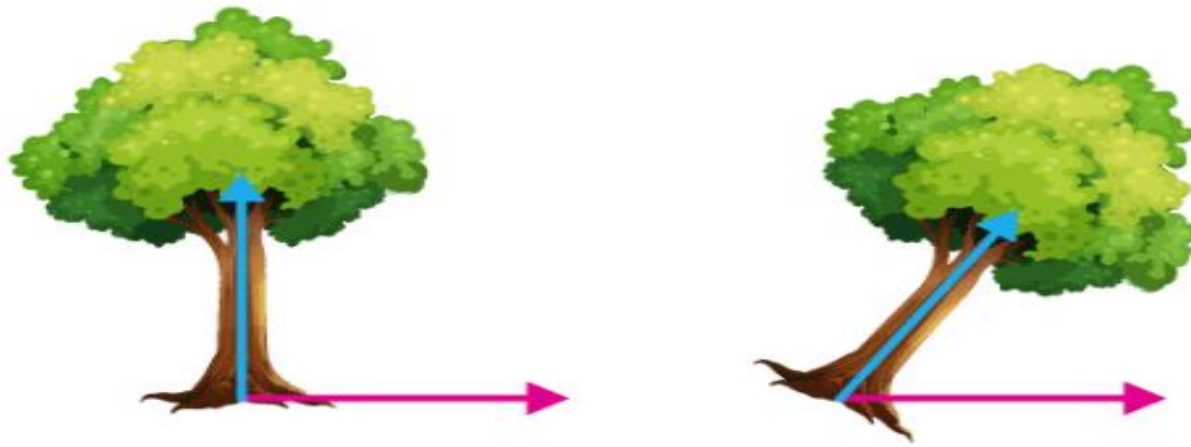
- **Covariance** is used to measure variables that have different units of measurement. So covariance does not use one standardized unit of measurement.
- **Correlation**, on the other hand, standardizes the measure of interdependence between two variables and informs researchers as to how closely the two variables move together.

Concept of Feature Preprocessing of data:

- **Eigenvalue** - the eigenvalue is a scalar that is used to transform the eigenvector.
- The basic equation is $\mathbf{Ax} = \lambda\mathbf{x}$
- The number or scalar value “ λ ” is an eigenvalue of A.

EigenValue Example

In this shear mapping, the blue arrow changes direction, whereas the pink arrow does not. Here, the pink arrow is an eigenvector because it does not change direction. Also, the length of this arrow is not changed; its eigenvalue is 1.



Principal Component Analysis

- Hence, the
- two eigenvalues
- of the given
- matrix are
- $\lambda = 0$ and $\lambda = 4$.
- Eigen matrix
- can be written
- as,
- $|A - \lambda I|$
- and
- $|A - \lambda I| = 0$

Find the eigenvalues of the 2 x 2 matrix

$$A = \begin{bmatrix} 0 & -2 \\ 3 & 4 \end{bmatrix}$$

Solution:

Given,

$$A = \begin{bmatrix} 0 & -2 \\ 3 & 4 \end{bmatrix}$$

Using the characteristic equation,

Let

$$I = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

be the 2 x 2 identity matrix.

$$|A - \lambda I| = 0$$

$$\left| \begin{bmatrix} 0 & -2 \\ 3 & 4 \end{bmatrix} - \lambda \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \right| = 0$$

$$\left| \begin{bmatrix} 0 & -2 \\ 3 & 4 \end{bmatrix} - \begin{bmatrix} \lambda & 0 \\ 0 & \lambda \end{bmatrix} \right| = 0$$

$$\begin{vmatrix} -\lambda & -2 \\ 0 & 4 - \lambda \end{vmatrix} = 0$$

$$-\lambda(4 - \lambda) - (-2)(0) = 0$$

$$-4\lambda + \lambda^2 = 0$$

$$\lambda(\lambda - 4) = 0$$

$$\lambda = 0 \text{ or } \lambda - 4 = 0$$

$$\text{Thus, } \lambda = 0 \text{ or } \lambda = 4$$

Principal Component Analysis

- **Eigenvector**- Eigenvector of a square matrix is defined as a non-vector in which when a given matrix is multiplied.
- If A be an $n \times n$ matrix and λ be the eigenvalues associated with it.
- Then, eigenvector v can be defined by the following relation: **$Av = \lambda v$**

Principal Component Analysis

Eigenvector Examples

Example: Find the eigenvector of the given matrix:

$$A = \begin{bmatrix} 1 & 4 \\ -4 & -7 \end{bmatrix}$$

Solution:

Given:

$$A = \begin{bmatrix} 1 & 4 \\ -4 & -7 \end{bmatrix}$$

$$|A - \lambda I| = \begin{vmatrix} 1 - \lambda & 4 \\ -4 & -7 - \lambda \end{vmatrix}$$

$$(1 - \lambda)(-7 - \lambda) - 4(-4) = 0$$

$$(\lambda + 3)^2 = 0$$

Therefore, $\lambda = -3, -3$

Use the eigenvector equation

$$AX = \lambda X$$

Substitute λ value in the equation:

$$AX = -3X$$

We know that,

$$(A - \lambda I)X = 0$$

$$\left(\begin{bmatrix} 1 & 4 \\ -4 & -7 \end{bmatrix} + \begin{bmatrix} 3 & 0 \\ 0 & 3 \end{bmatrix} \right) \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$4x + 4y = 0$$

Or

$$x + y = 0$$

Assume that $x = k$

So, it becomes

$$k + y = 0$$

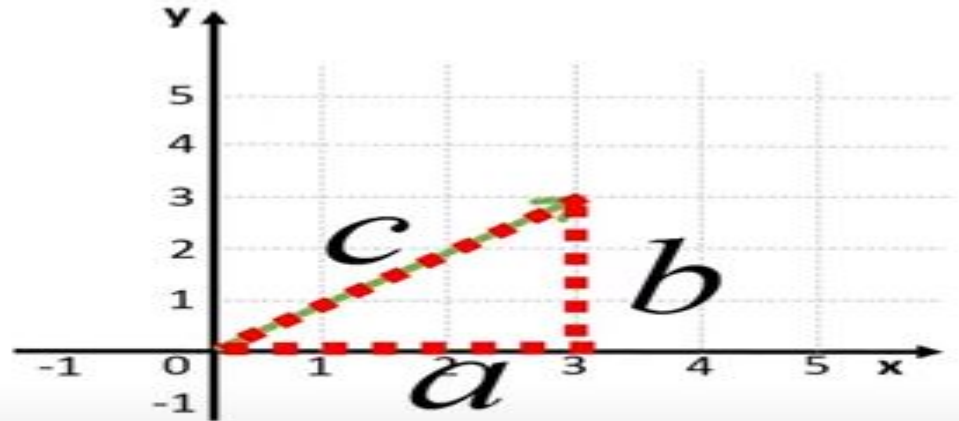
$$y = -k$$

Therefore, the eigenvector is

$$X = \begin{bmatrix} x \\ y \end{bmatrix} = k \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

Principal Component Analysis

$$v = \begin{bmatrix} 3 \\ 3 \end{bmatrix}$$

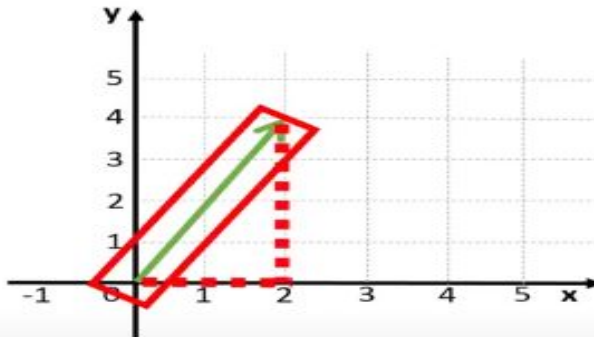


Pythagoras' theorem

$$c^2 = a^2 + b^2$$

Vectors

$$v = \begin{bmatrix} 2 \\ 4 \end{bmatrix}$$



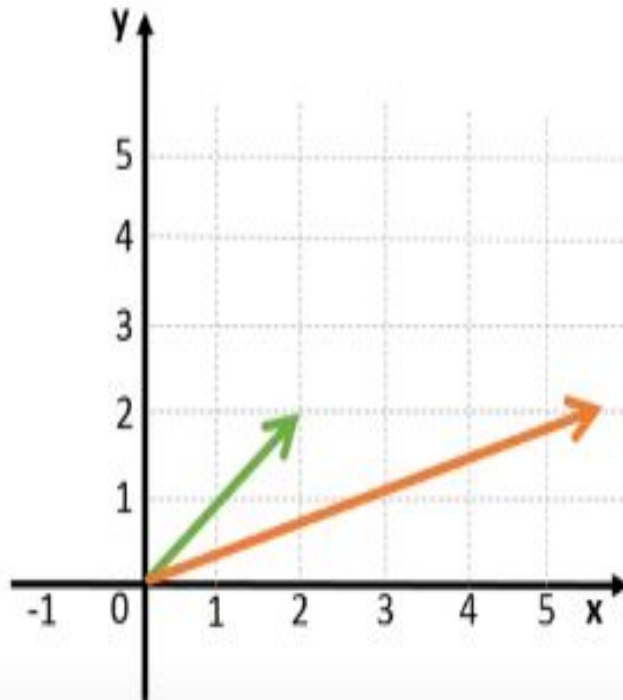
$$c = \sqrt{2^2 + 4^2} = \sqrt{20}$$

The length of this vector is the square root of 20, since the length of the two dashed lines of this triangle are two and four.

Eigenvectors

$$v = \begin{bmatrix} 2 \\ 2 \end{bmatrix}$$

$$A = \begin{bmatrix} 1 & 2 \\ 1 & 0 \end{bmatrix}$$



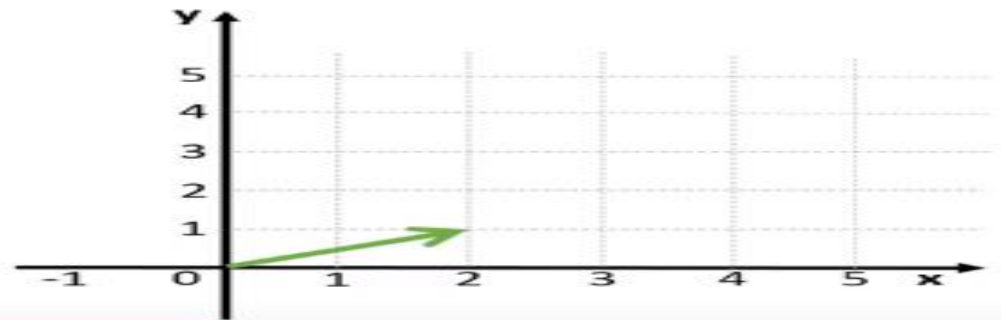
$$Av = \begin{bmatrix} 1 & 2 \\ 1 & 0 \end{bmatrix} \cdot \begin{bmatrix} 2 \\ 2 \end{bmatrix} = \begin{bmatrix} 6 \\ 2 \end{bmatrix}$$

Since this vector changes its direction when multiplied with matrix A, we can conclude that this vector is not an eigenvector of matrix A.

Eigenvectors

$$v = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$$

$$A = \begin{bmatrix} 1 & 2 \\ 1 & 0 \end{bmatrix}$$

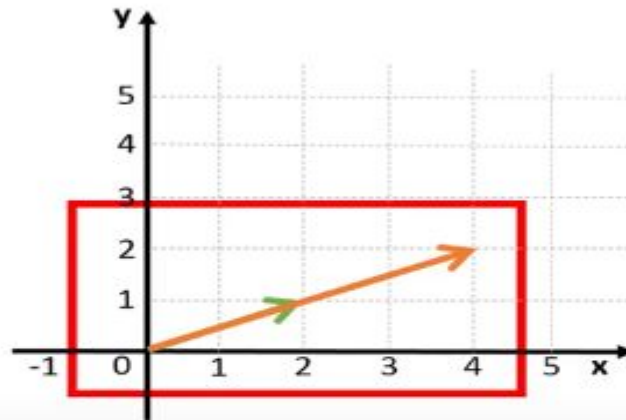


$$Av = \begin{bmatrix} 1 & 2 \\ 1 & 0 \end{bmatrix} \cdot \begin{bmatrix} 2 \\ 1 \end{bmatrix} = \begin{bmatrix} 4 \\ 2 \end{bmatrix}$$

Eigenvalue

$$v = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$$

$$A = \begin{bmatrix} 1 & 2 \\ 1 & 0 \end{bmatrix}$$



$$Av = \begin{bmatrix} 1 & 2 \\ 1 & 0 \end{bmatrix} \cdot \begin{bmatrix} 2 \\ 1 \end{bmatrix} = \begin{bmatrix} 4 \\ 2 \end{bmatrix}$$

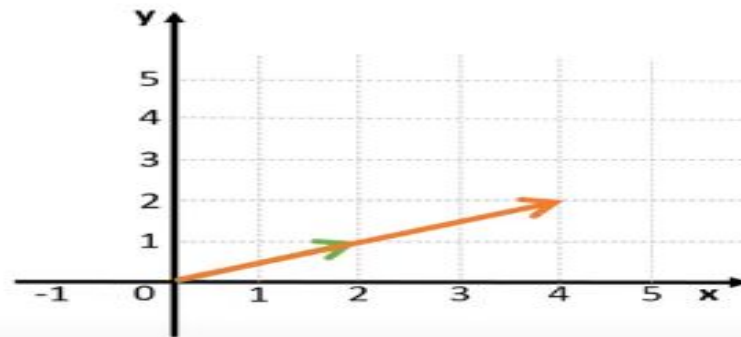
If we draw this vector in the plane, we see that the new vector has the same direction as the original vector. Thus, vector v is therefore an eigenvector of matrix A .

Eigenvalue

$$v = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$$

$$A = \begin{bmatrix} 1 & 2 \\ 1 & 0 \end{bmatrix}$$

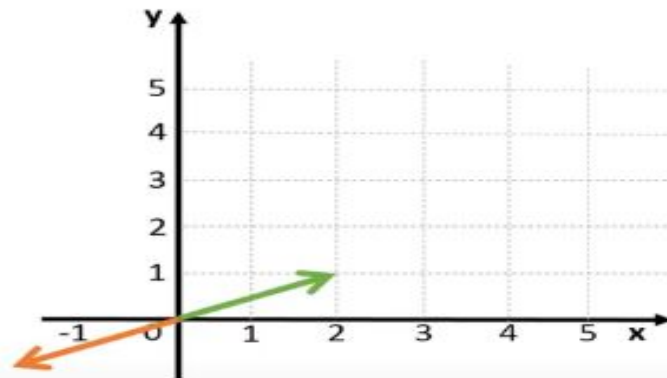
$$Av = \begin{bmatrix} 4 \\ 2 \end{bmatrix} = 2 \begin{bmatrix} 2 \\ 1 \end{bmatrix}$$



$$\lambda = 2$$

two is the eigenvalue,

Eigenvalue



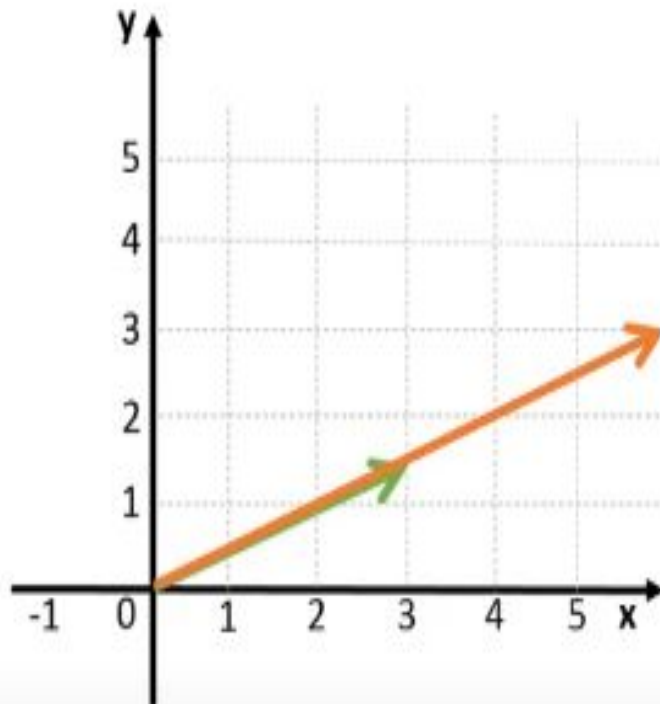
$$Av = \lambda v$$

Note that, the eigenvalue can even be negative, which means that the direction of the new vector is reversed but still on the same "line".

Eigenvectors

$$v = \begin{bmatrix} 3 \\ 1.5 \end{bmatrix}$$

$$A = \begin{bmatrix} 1 & 2 \\ 1 & 0 \end{bmatrix}$$



$$Av = \begin{bmatrix} 1 & 2 \\ 1 & 0 \end{bmatrix} \cdot \begin{bmatrix} 3 \\ 1.5 \end{bmatrix} = \begin{bmatrix} 6 \\ 3 \end{bmatrix} = 2 \begin{bmatrix} 3 \\ 1.5 \end{bmatrix}$$

Note that, although we have found a new eigenvector, the corresponding eigenvalue does not change. We see that the eigenvalue is still 2.

Principal Component Analysis

Advantages of Dimensionality Reduction

- It helps in data compression, and hence reduced storage space.
- It reduces computation time.
- It also helps remove redundant features, if any.

Disadvantages of Dimensionality Reduction

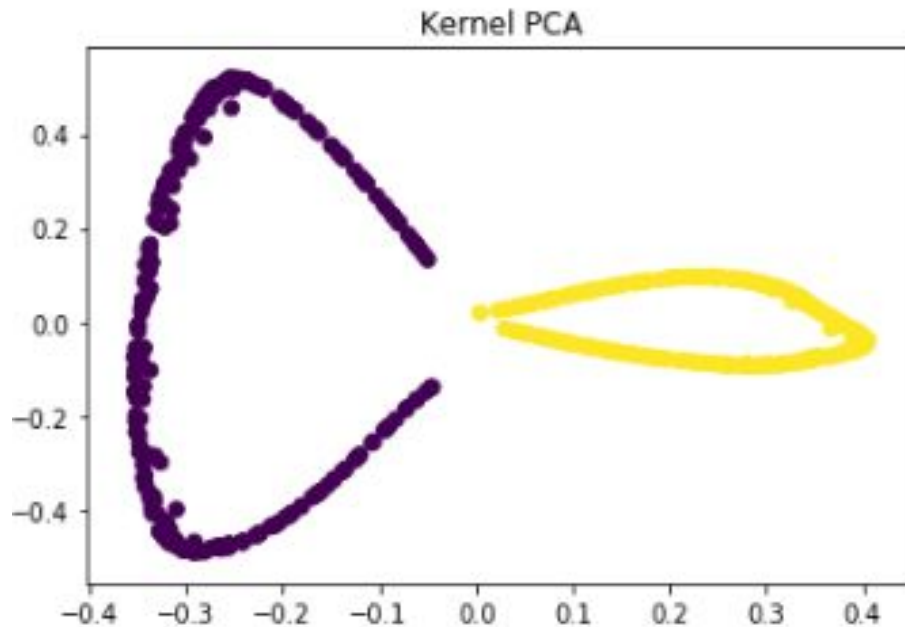
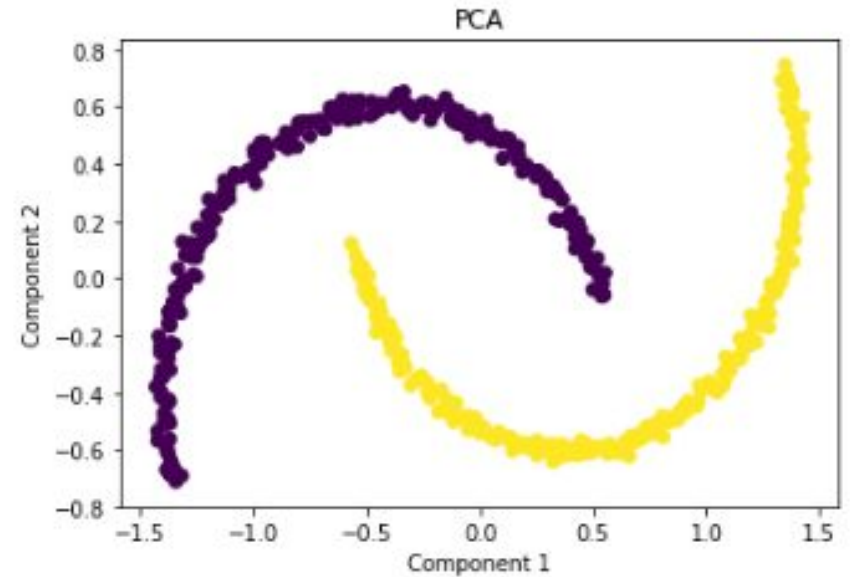
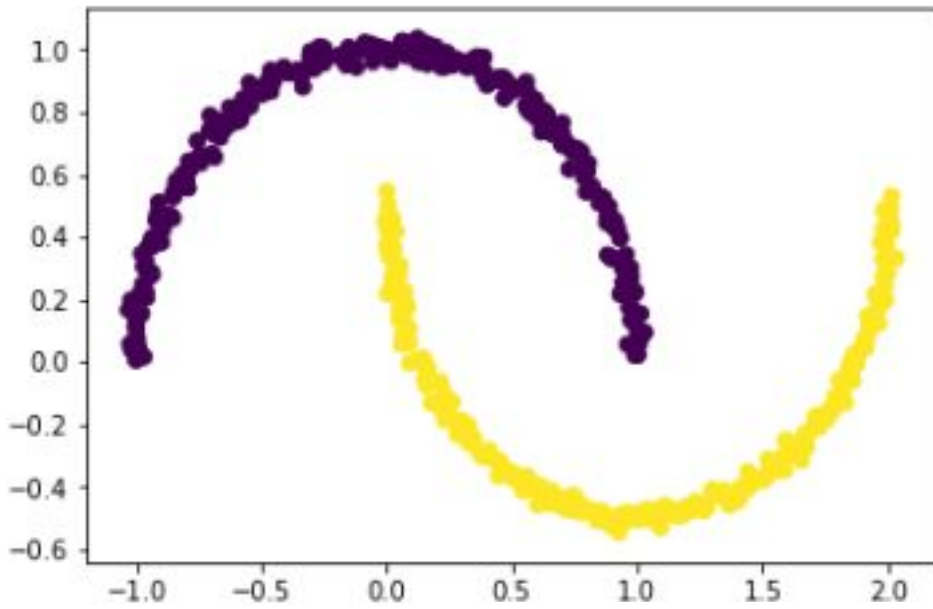
- It may lead to some amount of data loss.
- PCA tends to find linear correlations between variables, which is sometimes undesirable.
- PCA fails in cases where mean and covariance are not enough to define datasets.
- We may not know how many principal components to keep- in practice, some thumb rules are applied.

- **Some common terms used in PCA algorithm:**
- **Dimensionality:** It is the number of features or variables present in the given dataset. More easily, it is the number of columns present in the dataset.
- **Correlation:** It signifies that how strongly two variables are related to each other. Such as if one changes, the other variable also gets changed. The correlation value ranges from -1 to +1. Here, -1 occurs if variables are inversely proportional to each other, and +1 indicates that variables are directly proportional to each other.
- **Orthogonal:** It defines that variables are not correlated to each other, and hence the correlation between the pair of variables is zero.
- **Eigenvectors:** If there is a square matrix M , and a non-zero vector v is given. Then v will be eigenvector if Av is the scalar multiple of v .
- **Covariance Matrix:** A matrix containing the covariance between the pair of variables is called the Covariance Matrix.

Kernel PCA

- PCA is a linear method. That is it can only be applied to datasets which are linearly separable. It does an excellent job for datasets, which are linearly separable.
- But, if we use it to non-linear datasets, we might get a result which may not be the optimal dimensionality reduction.
- Kernel PCA uses a kernel function to project dataset into a higher dimensional feature space, where it is linearly separable. It is similar to the idea of Support Vector Machines.
- There are various kernel methods like linear, polynomial, and Gaussian.
- In the kernel space the two classes are linearly separable. **Kernel PCA uses a kernel function to project the dataset into a higher-dimensional space, where it is linearly separable.**

Kernel PCA



Local Binary Pattern

- is in its ability to differentiate tiny differences in texture and topography, to identify key features with which we can then differentiate between images of the same type — no painstaking labeling required.
- **The goal of LBP is to encode geometric features of an image by detecting edges, corners, raised or flat areas and hard lines; allowing us to generate a feature vector representation of an image, or group of images.**

Local Binary Pattern

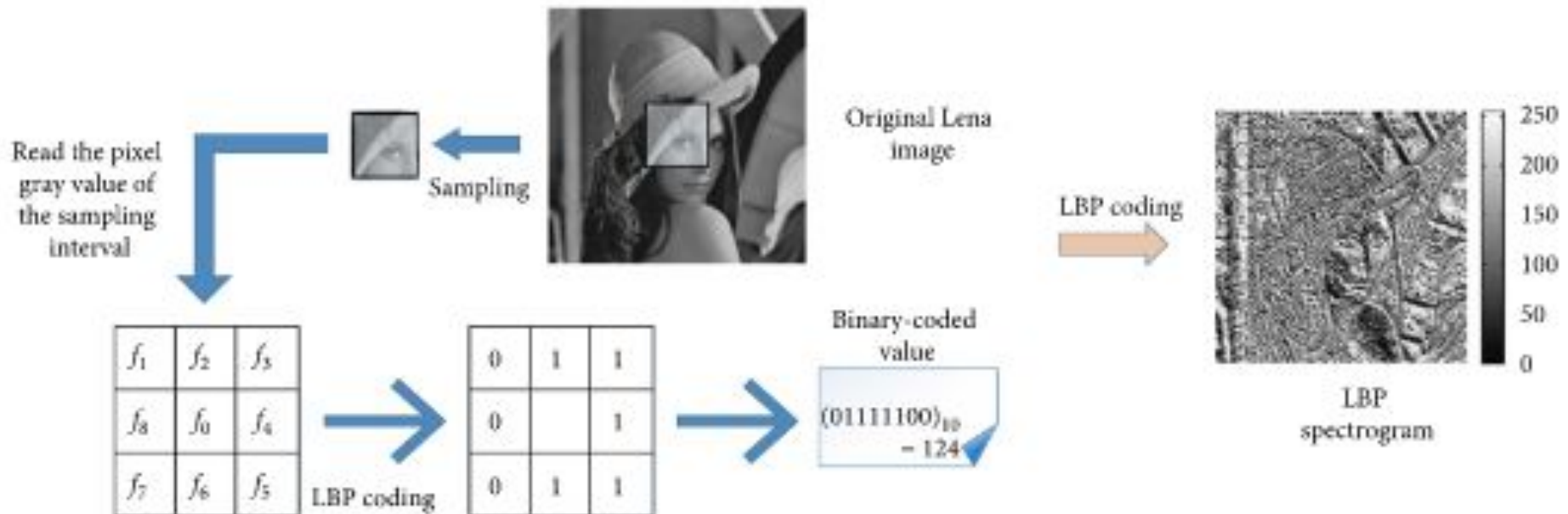
- we can determine the level of similarity between our target representation and an unseen image and can **calculate the probability that the image presented** is of the same variety or type as the target image.
- **LBP can be split into 4 key steps:**
 - Simplification
 - Binarisation
 - PDF (probability density function) calculation
 - Comparison (of the above functions)
- *Simplification*

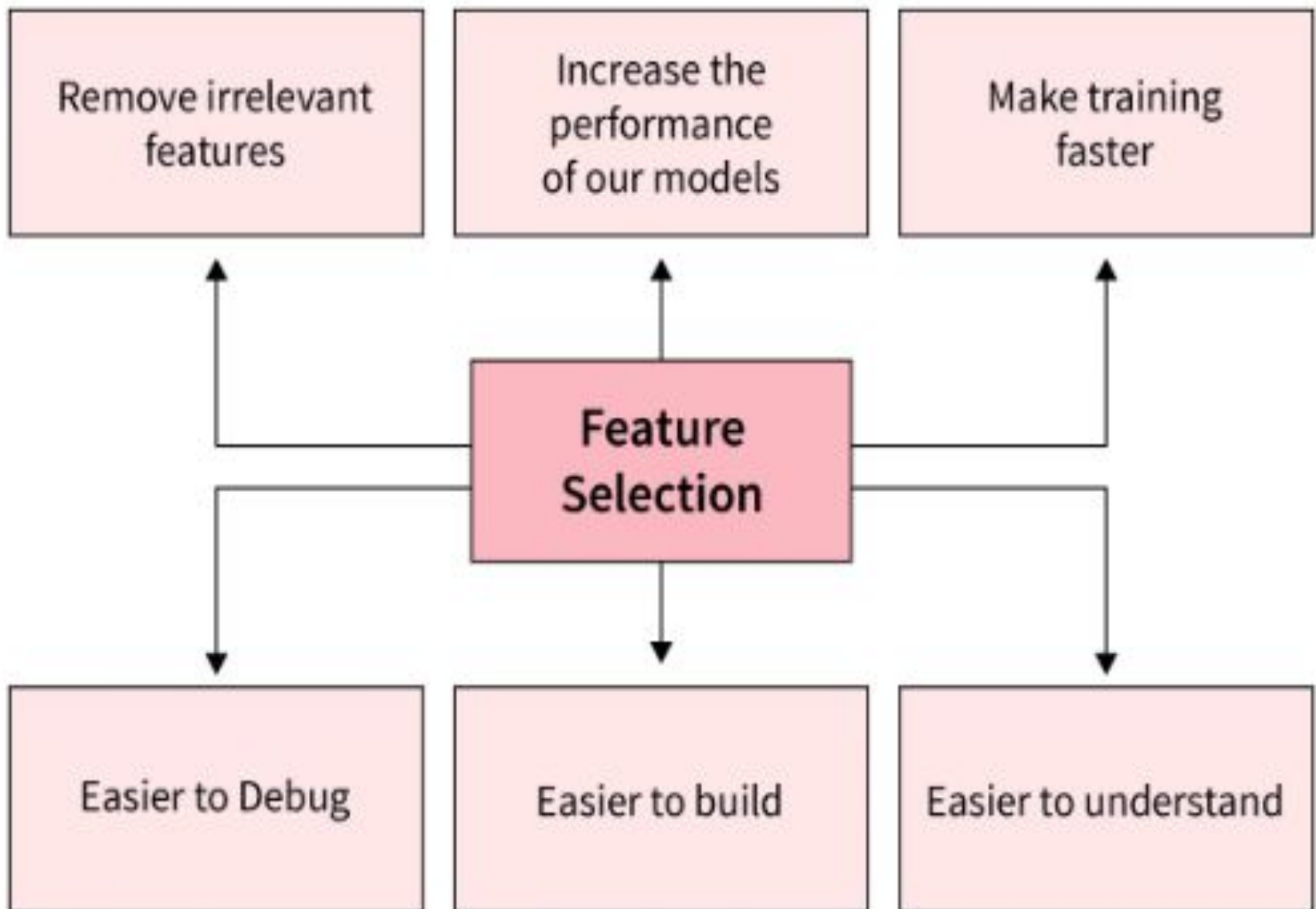
Local Binary Pattern

- This is our data preprocessing step. In essence, this is our first step in dimensionality reduction, which allows our algorithm to focus purely on the local differences in luminance, rather than worrying about any other potential features.
- Therefore, we first convert our image into a single channel (typically greyscale) representation
- **Binarisation**
- Next, we calculate the relative local luminance changes. **This allows us to create a local, low dimensional, binary representation of each pixel based on luminance.**

Local Binary Pattern

- For each comparison, we output a binary value of 0 or 1, dependent on whether the central pixel's intensity (scalar value) is greater or less (respectively) than the comparison pixel.
- This forms a k -bit binary value, which can then be converted to a base 10 number; forming a new intensity for that given pixel.





Selection vs. Extraction

- In **feature selection** we try to find the best subset of the input feature set.
- In **feature extraction** we create new features based on transformation or combination of the original feature set.
- Both selection and extraction lead to the dimensionality reduction.
- No clear cut evidence that one of them is superior to the other on all types of task.

There are two types of feature selection techniques in Machine Learning - **Supervised Feature Selection technique** considers the target variable and can be used for the labeled dataset.

Unsupervised Feature Selection technique ignores the target variable and can be used for the unlabelled dataset.

Let's see different methods for each of these categories.

Feature Selection Techniques

Supervised Feature Selection

Unsupervised Feature Selection

Filters method

- Missing value
- Information gain
- Chi-square Test
- Fisher's Score

Embedded method

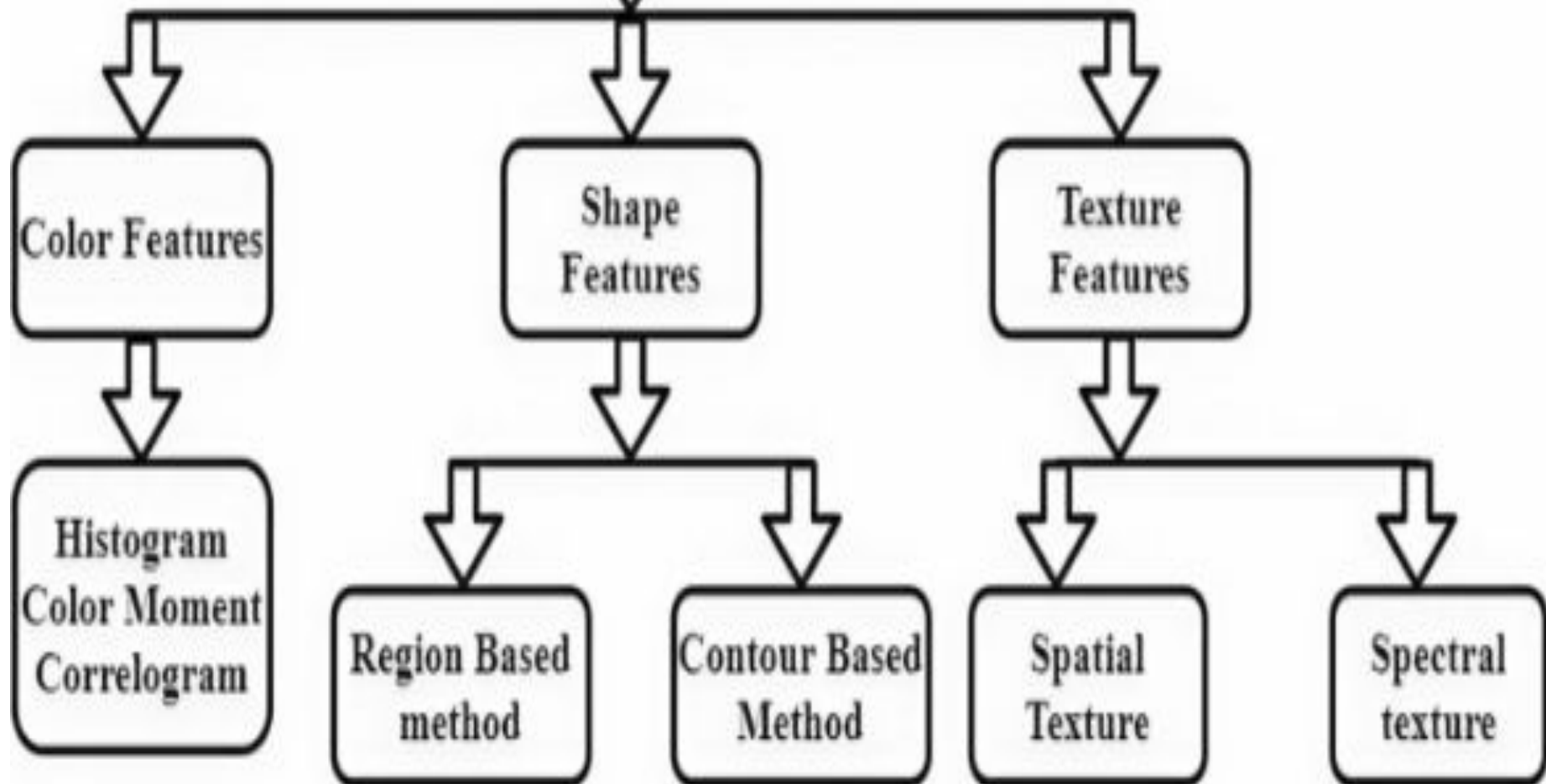
- Regularization L1, L2
- Random forest Importance

Wrappers method

- Forward Feature Selection
- Backward Feature Selection
- Exhaustive Feature Selection
- Recursive Feature Selection



Feature
Extraction



Sequential Forward Selection

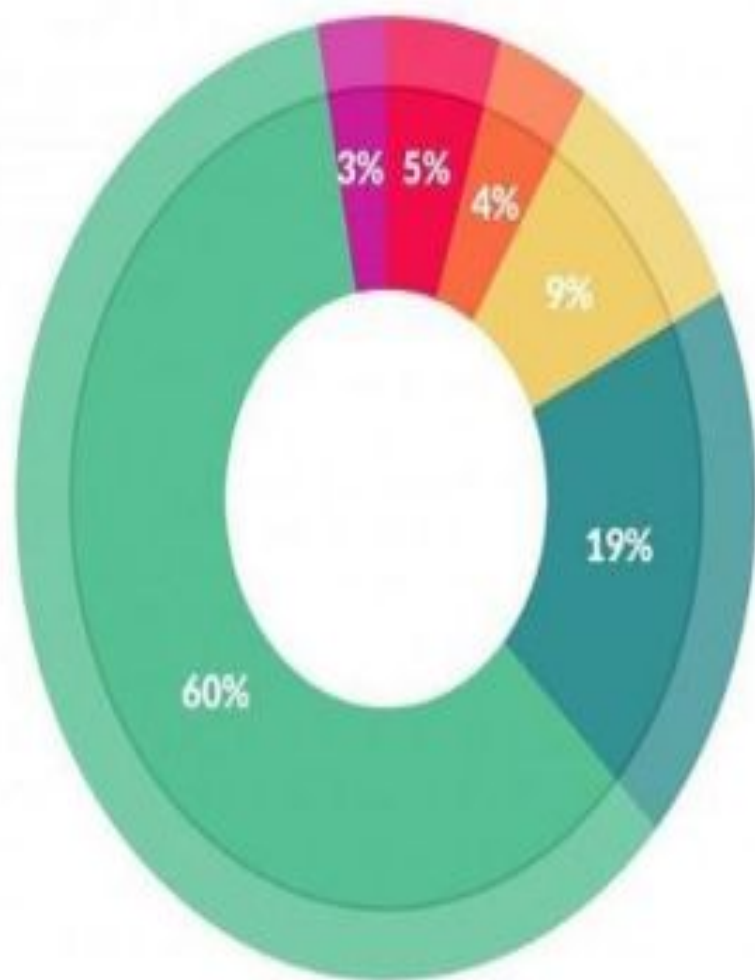
- SFS
- At the beginning select the best feature using a scalar criterion function.
- Add one feature at a time which along with already selected features maximizes the criterion function.
- A **greedy** algorithm, cannot retract (also called nesting effect).

Sequential Backward Selection

- SBS
- At the beginning select all d features.
- Delete one feature at a time and select the subset which maximize the criterion function.
- Also a greedy algorithm, cannot retract.
- Complexity is $O(d)$.

Statistical feature engineering:

- **Feature engineering** refers to a process of **selecting & transforming** variables/features in your dataset when creating a **predictive model** using machine learning.
- Therefore you have to extract the features from the **raw dataset** you have collected before training your data in machine learning algorithms.
- Feature engineering has two goals:
- Preparing the proper input dataset, compatible with the machine learning algorithm requirements.
- Improving the **performance** of machine learning models.



What data scientists spend the most time doing

- Building training sets: 3%
- Cleaning and organizing data: 60%
- Collecting data sets: 19%
- Mining data for patterns: 9%
- Refining algorithms: 4%
- Other: 5%

CrowdFlower Survey

Statistical feature engineering:

- **Mean:** The "average" number; found by adding all data points and dividing by the number of data points.
- Example: The mean of 444, 111, and 777 is $(4+1+7)/3 = 12/3 = 4$ left parenthesis, 4, plus, 1, plus, 7, right parenthesis, slash, 3, equals, 12, slash, 3, equals, 4.

Statistical feature engineering:

- **Median:** The middle number; found by ordering all data points and picking out the one in the middle (or if there are two middle numbers, taking the mean of those two numbers).
- **Example:** The median of 444, 111, and 777 is 444 because when the numbers are put in order (1(left parenthesis, 1, 444, 7)7)7, right parenthesis, the number 444 is in the middle.

Statistical feature engineering:

- **Mode:** The most frequent number—that is, the number that occurs the highest number of times.
- Example: The mode of $\{4, 222, 444, 333, 222, 2\}$, right brace is 222 because it occurs three times, which is more than any other number.

Feature Vector Creation.

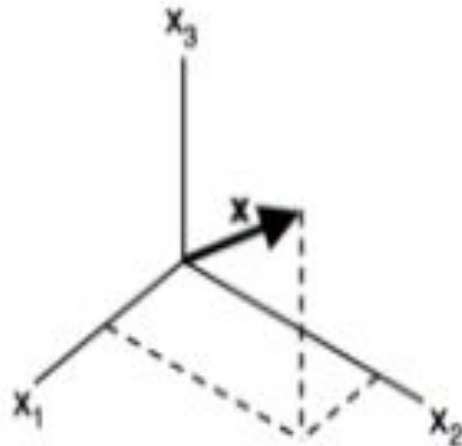
- A vector is a series of numbers, like a matrix with one column but multiple rows, that can often be represented spatially.
- A **feature** is a numerical or symbolic property of an aspect of an object.
- A **feature vector** is a vector containing **multiple elements about an object**. Putting feature vectors for objects together can make up a **feature space**.

Statistical feature engineering:

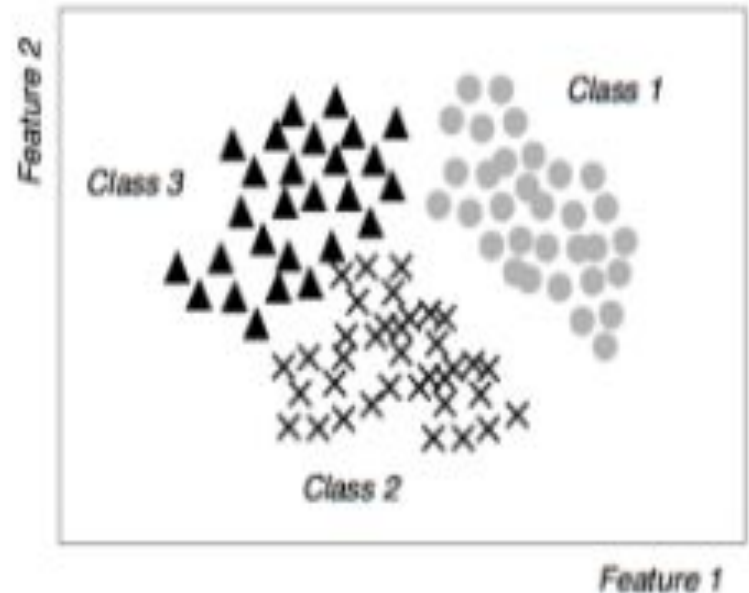
- The features may represent, as a whole, one mere pixel or an entire image. The granularity depends on what someone is trying to learn or represent about the object. You could describe a 3-dimensional shape with a feature vector indicating its height, width, depth, etc.

$$\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_d \end{bmatrix}$$

Feature vector



Feature space (3D)



Scatter plot (2D)

Statistical feature engineering:

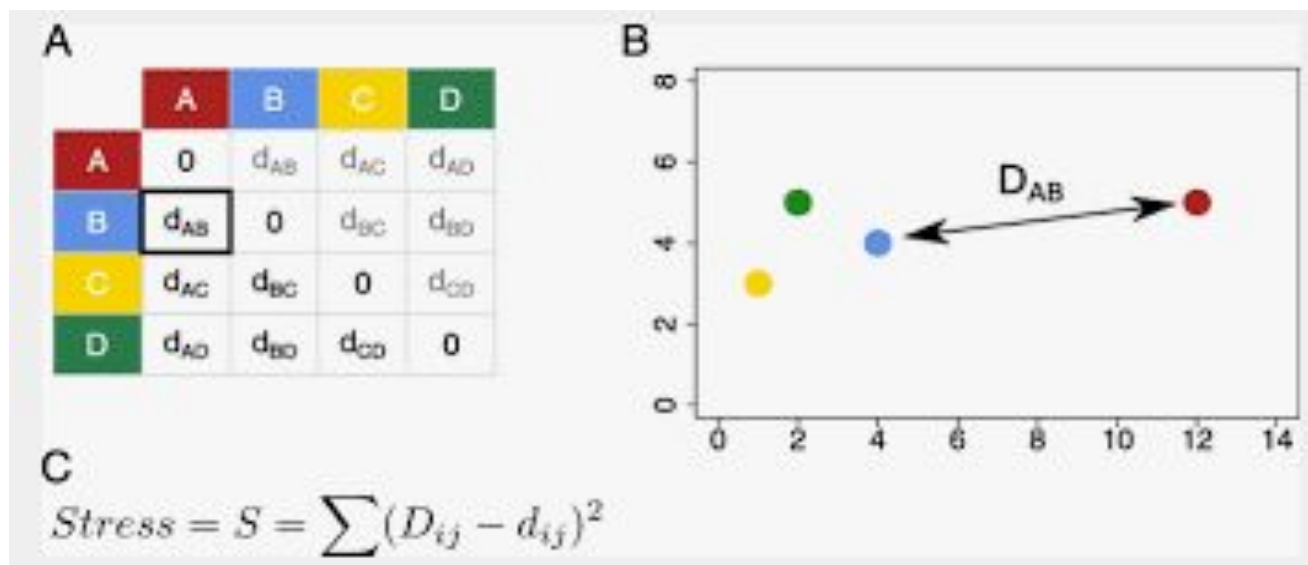
- One simple way to compare the feature vectors of two objects is to take the Euclidean distance.
- In **image processing**, features can be gradient magnitude, color, grayscale intensity, edges, areas, and more.
- Feature vectors are particularly popular for analyses in image processing because of the convenient way attributes about an image, like the examples listed, can be compared numerically once put into feature vectors.
- In **speech recognition**, features can be sound lengths, noise level, noise ratios, and more.

- In **spam-fighting initiatives**, features are abundant. They can be IP location, text structure, frequency of certain words, or certain email headers.
- Feature vectors are used in [classification problems](#), [artificial neural networks](#), and [k-nearest neighbors](#) algorithms in machine learning.
- **Feature vector is an n-dimensional vector of numerical features that describe some object in pattern recognition in machine learning**

What is Multidimensional Scaling?

- Multidimensional scaling is a **visual representation of distances or dissimilarities between of high dimensional data.**
- The “**multidimensional**” part is due to the fact that you aren’t limited to **two dimensional graphs or data. Three-dimensional, four-dimensional and higher plots are possible.**
- Its use in geostatistics helps visually assess and understand multivariate data in a lower dimension.

- By reducing the dimensionality of the data one can observe patterns, gradients, and clusters that may be helpful in exploratory data analysis.
- MDS does this by projecting the multivariate distances between entities to a best-fit configuration in lower dimensions.



Model listening data as a product of latent factors

